

SEMANTIC WEB PROJECT

VN-Edu LOD

A Knowledge Graph of Vietnamese Higher-Education Institutions

Design, Implementation, and Evidence-based Evaluation

Revised English project report

Ontology 2.2 and integration edition: 8 October 2026

This report presents the design and implementation of a Linked Open Data knowledge graph for Vietnamese higher-education institutions. It combines a domain ontology, a reproducible multi-source transformation pipeline, RDF publication, reasoning, validation, and SPARQL access. The report is based on the corrected repository release and reports measured results with their scope, rather than treating a successful query or a visualization as proof of real-world truth.

Project repository:

<https://github.com/pham-ng/Vietnam-University-Knowledge-Graph-ver2>

Abstract

VN-Edu LOD is an RDF knowledge graph for Vietnamese educational institutions. The project integrates structured Wikidata records, Vietnamese Wikipedia material, and curated reference tables into a common model of organizations, people, academic programs, and administrative geography. The corrected snapshot contains 300 educational organizations, 1,702 people, 34 current and 29 former province-level units, and 78 illustrative programs. Its central technical contribution is a staged Bronze–Silver–Gold pipeline: source snapshots are normalized into records, records are mapped to persistent HTTP IRIs, RDF is linked and materialized, and the resulting union is released only after structural and semantic checks.

The ontology contains 39 local classes, 38 object properties, and 27 datatype properties. OWL 2 RL profile checking, rule-based materialization, SHACL validation, Apache Jena Fuseki/TDB2 acceptance tests, and regression tests provide complementary evidence. The validated serving union contains 73,628 distinct triples. The report separates ontology conformance, reference-link agreement, data completeness, factual accuracy, and production readiness: they are related engineering goals, but none is a substitute for the others. The result is a reproducible educational Linked Data artifact with an explicit list of remaining evidence requirements for authoritative, high-load deployment.

Keywords: knowledge graph; RDF; ontology; SHACL; SPARQL; entity resolution; data provenance; Vietnamese higher education.

Contents

1	Project overview	6
1.1	Motivation and problem statement	6
1.2	Objectives and contributions	6
1.3	Research and evaluation questions	7
2	Semantic Web foundations and requirements	7
2.1	Standards used by the project	7
3	Data sources and collection	8
3.1	Source roles	8
3.2	Scope and selection effects	9
4	Architecture and data management	9
4.1	Layers and contracts	9
4.2	Identity and URI persistence	10
4.3	Provenance and disputed values	10
5	Transformation: from source records to RDF	11
5.1	Integration policies and information loss	11
5.2	Concrete mapping example	12
5.3	Linking, reasoning and release boundaries	12
6	Ontology design and relationship semantics	13
6.1	Design method	13
6.2	Modules	13
6.3	Visual specification of ontology 2.2	14
6.4	TBox, ABox and constraint semantics	25
6.5	Implemented corrections and evidence limits	25
6.6	Standards and conformance assessment	26
6.7	Qualified observations and temporal interpretation	27
6.8	Competency questions and prohibited conclusions	27
6.9	Materialization	30
6.10	Geographic and temporal limitations	30
7	Linking and publication	31
7.1	External relations	31
7.2	Representations and deployment	31
8	Deployment and Apache Jena Fuseki	32

8.1	Identity, representation and storage	32
8.2	Safe loading and reproducible runtime	32
8.3	Production-like security and performance acceptance	33
8.4	Production hardening profile and release gates	33
8.5	Executed end-to-end acceptance evidence	34
9	Five-star Linked Open Data assessment	35
9.1	Dated HTTP evidence	35
10	Application, security and CI/CD	36
11	Evaluation and results	37
11.1	Release measurements	37
11.2	Link experiment: agreement, not independent accuracy	37
11.3	Executed Silk Framework experiment	38
11.4	Regression evidence	39
11.5	Treatment of earlier comparisons	39
12	Discussion, limitations and future work	40
13	Conclusion	40
A	Reproducing the release and report	41
B	A competency-query example	42

List of Figures

1	Validated release flow	10
2	Transformation stages and data artifacts	11
3	Main unique-node ontology map	14
4	Protégé OntoGraf view of ontology 2.2	15
5	Protégé object-property signature	16
6	Complete local TBox class hierarchy	19
7	Organization and governance signatures	20
8	Education and people signatures	21
9	Geography and change signatures	22
10	Other local object-property signature	23
11	ABox instance neighborhood for a real university	24
12	Organizational class hierarchy	28
13	Property signatures across ontology modules	28
14	Geographic property-chain inference	29
15	Evidence-qualified temporal observations	29
16	URI publication and optional Fuseki execution	32

List of Tables

1	Project contributions and evidence.	6
2	Design requirements and the mechanism used to address them.	8
3	Source roles and integration treatment.	8
4	Selected Silver-to-RDF mappings for a real input record.	12
5	Main modules and intended distinctions.	13
6	Separation of ontology axioms, data assertions and validation constraints.	25
7	Ontology 2.2 corrections and their limits.	25
8	Standards assessment and remaining boundaries.	26
9	Technology roles in the VN-Edu publication architecture.	31
10	Production release gates that remain separate from OWL and SHACL conformance. . .	34
11	Measured Fuseki/TDB2 acceptance results.	35
12	Evidence against the cumulative five-star publication scheme.	35
13	Observed HTTP behavior; deployed content may predate the corrected branch.	36
14	CI/CD gates and the claim supported by each gate.	37
15	Measured size of the corrected local release.	38
16	Token-Jaccard agreement with the identifier-derived reference.	38
17	Actual Silk versus the fixed-sample Python baseline: reference agreement only.	39

1 Project overview

1.1 Motivation and problem statement

Information about a university is distributed across knowledge bases, encyclopedic articles, institutional pages, and administrative reference data. The same institution may have several names, identifiers, locations, or organizational relationships. These differences matter in practice: a member school is not necessarily a branch campus, direct governance is not the same as state oversight, and a former province is not interchangeable with its successor. An RDF knowledge graph is useful here because it represents those distinctions as typed, linkable statements that can be queried across sources.

VN-Edu LOD is an engineering case study of this integration problem. It is designed to demonstrate a complete Semantic Web workflow, from source collection and URI assignment to ontology-based transformation, reasoning, validation, publication, and query execution. It is not presented as a complete official register of Vietnamese higher education. Names, classifications, dates, and relationships describe the collected snapshot and the project’s documented extraction policies.

1.2 Objectives and contributions

The project has four practical objectives:

1. build a domain ontology that distinguishes organization, governance, people, education, geography, identity, and temporal evidence;
2. transform heterogeneous records into persistent RDF resources and publish machine-readable representations using Linked Data principles;
3. evaluate the release with OWL 2 RL profile checking, materialization, SHACL, SPARQL, and automated regression tests; and
4. demonstrate reproducible execution with Apache Jena Fuseki and a controlled Silk candidate-linking experiment.

The resulting contributions are summarized in Table 1. This framing follows a project-report structure: each contribution is paired with an artifact and an evidence source, so that the report can distinguish implementation from evaluation.

Table 1: Project contributions and evidence.

Contribution	Implemented artifact	Evidence reported in this document
Domain model	OWL/Turtle ontology 2.2 with classes, properties, signatures, alignments, and selected chains	Protégé inspection, OWL 2 RL profile gate, ontology figures, and reasoning examples
Data integration	Bronze–Silver–Gold transformation pipeline, URI registry, provenance and conflict logs	Reproducible pipeline, mapping example, release fingerprints, and coverage counts
Linked publication	HTML, Turtle, JSON-LD, stable resource IRIs, link graph, and metadata	Publication checks, five-star assessment, and explicit licensing/identity scope
Operational checks	Flask read-only query path, Fuseki/TDB2 loader, CI workflows, SHACL and regression suite	End-to-end acceptance tables, security probes, and test results

1.3 Research and evaluation questions

The evaluation is organized around four questions:

1. Does the model distinguish the relationships needed for institutional, educational, geographic, and governance queries without producing detected local contradictions?
2. Can the transformation reproduce a validated release while preserving resource identity when labels or input order change?
3. What do the measurements establish about structure, coverage, link agreement, and factual correctness, and which claims remain unsupported?
4. Can publication and query handling fail safely on stale releases, invalid input, unapproved federation, and bounded overload?

The review covers the source code, ontology, SHACL shapes, cached inputs, generated RDF, tests, CI workflows, and earlier project documentation. Security counterexamples are executed locally; external production endpoints are not used as attack targets.

2 Semantic Web foundations and requirements

2.1 Standards used by the project

RDF describes resources through triples. Subjects may be IRIs or blank nodes; predicates are IRIs, and objects may additionally be literals. A graph is a set, so duplicate triples do not increase its size. This matters when combining asserted data, ontology statements, links, and inferred statements: their file counts cannot simply be added to obtain the size of the union [1].

OWL axioms express implications about the modeled world. A property's domain is not an input-validation rule: using the property can entail that its subject belongs to that domain. Similarly, a functional object property can imply equality between two values; it does not necessarily reject the second value. The project therefore supplements reasoning with explicit validation of asserted records. The OWL 2 profile specification defines restrictions under which particular reasoning guarantees apply [2].

SHACL validates a supplied data graph against declared constraints [3]. Conformance is relative to those constraints, not a claim that every statement is true. Here, missing values for some descriptive fields generate warnings, while invalid datatypes, selected cardinality violations, and hard constraints prevent publication.

SPARQL supplies query syntax and algebra for RDF, including optional patterns, aggregates, and remote SERVICE clauses [4]. Exposing it over HTTP introduces an execution boundary: query syntax, remote destinations, elapsed work, and publication state must all be controlled independently of the user interface.

The project reuses identifiers and vocabulary from Wikidata, DBpedia, schema.org, FOAF, SKOS, and PROV-O. PROV-O supports descriptions of derivation [5]; whether the provenance is attached to an entity, a graph, or an individual assertion determines what can actually be traced. HTTP identifiers and useful representations support Linked Data publication, while persistence requires more than generating readable slugs [6].

The standards are assigned different responsibilities. RDF and HTTP IRIs provide the data model and Web identity; OWL provides class and property semantics; SHACL provides explicit validation constraints; SPARQL provides query access; and Fuseki provides a storage/query runtime. Silk is used only for an experimental candidate-linking task. This separation is important because a tool boundary

is not a semantic guarantee: loading RDF into Fuseki does not mint IRIs, SHACL does not prove facts, and OWL profile membership does not certify data accuracy.

Table 2: Design requirements and the mechanism used to address them.

Requirement	Design response	Verification boundary
Interoperability	RDF 1.1, OWL, SKOS, PROV-O, JSON-LD, HTTP IRIs	Syntax and profile checks; external vocabulary semantics are not fully imported
Identity stability	Canonical resource namespace and committed URI registry	URI-regression tests; migration is still required when source keys change
Semantic precision	Separate properties for governance, oversight, membership, branch, ownership, and geography	Domain/range inspection and competency queries
Data quality	JSON Schema, raw asserted checks, OWL 2 RL materialization, and SHACL	Release gate; warnings remain visible and are not silently converted to facts
Reproducibility	Cached inputs, fingerprints, pinned runtime versions, CI workflows, and machine-readable reports	Local reconstruction and Linux Fuseki/Silk workflows

3 Data sources and collection

3.1 Source roles

Wikidata contributes identifiers, labels, institutional attributes, organizational links, and person-related records. Vietnamese Wikipedia contributes infobox fields, descriptions, historical text, and links between articles. DBpedia supplies candidate identity links and some fallback values. Selected reference tables describe province mergers, regional groupings, fields of study, majors, and example programs. Geocoding is used as a fallback for location resolution and is not equivalent to an authoritative campus register.

Table 3: Source roles and integration treatment.

Source or input	Information used	Treatment in the release
Wikidata	Identifiers, labels, types, links, and person/organization attributes	Primary identity evidence when a stable identifier is available; provenance is retained
Vietnamese Wikipedia	Infobox values, article descriptions, revisions, and page links	Structured extraction and revision-specific provenance; not treated as an official register
DBpedia and other linked datasets	Candidate external resources and identity links	Interlinking evidence; uncertain matches remain outside accepted identity assertions
Reference tables	Provinces, mergers, regions, fields, majors, and sample programs	Curated integration inputs; programs are illustrative rather than a complete catalogue
Geocoding fallback	Location resolution when structured identifiers are missing	Convenience support only; not evidence of legal campus status

Wikipedia and Wikidata are interconnected community resources. Agreement between them is useful corroborating evidence, but they must not be described as statistically independent sources. Imported statements can share an origin, and one resource may repeat an error from another. A disagreement log exposes selected conflicts; it does not establish which source is correct.

The collection strategy combines article-category traversal, infobox parsing, and Wikidata queries. Results are retained in JSON snapshots. The repository also ships a compressed HTTP cache. In the

revised default pipeline, a cache miss raises an error instead of silently contacting a changing remote source. A fresh collection is an explicit operation. Dependency installation may still require Internet access; the offline claim applies to cached data collection and reconstruction after dependencies are installed.

3.2 Scope and selection effects

The snapshot is a discovered set of educational organizations, not a census. Filtering removes unrelated articles and some records lacking sufficient cross-source evidence. That policy can exclude legitimate but poorly documented institutions. Institutions with more detailed encyclopedia coverage are more likely to have richer records. The same bias is particularly strong for people linked through education records.

The retained higher-education classification includes name-based heuristics. A title beginning with a familiar institution type is evidence used by the implementation; it is not a substitute for a current legal decision. Counts of higher-education institutions must therefore be read as counts under the project's classification policy.

Academic programs are illustrative hand-entered data. They demonstrate program-to-major queries but do not support claims about a university's complete or current admissions offerings. Similarly, a Wikidata education relation establishes recorded attendance or education, not necessarily completion of a degree. The existing *Alumnus* class must be interpreted with that limitation.

4 Architecture and data management

The implementation follows a layered architecture. Raw source material is kept separate from normalized records; RDF generation is separate from reasoning; and validated output is separate from publication. This organization is similar to the Bronze–Silver–Gold pattern used in other Linked Data projects, but the names describe project artifacts rather than a universal standard. The flow in Figure 1 is the release-level view; the detailed transformation diagram appears in Section 5.

4.1 Layers and contracts

Bronze stores responses and extraction snapshots. Silver contains institutions, people, governing bodies, and provinces in a normalized JSON representation. Gold contains asserted RDF, external links, selected instance-level entailments, and dataset metadata. The layers improve inspectability, but the implementation rewrites snapshot files during reconstruction. Bronze is therefore not an immutable append-only archive; Git revisions provide versioned snapshots.

Silver validation checks schema constraints and selected cross-field rules. The audit adds checks for future years, impossible calendar dates, missing cross-collection references, and invalid leadership references. These checks complement, rather than replace, semantic review of the extraction policy.

After RDF generation, literal ranges and functional-field cardinalities are validated on the raw asserted graph, independently of equality in the closure. The rule engine's own error messages are retained in the consistency diagnostics. SHACL then checks the asserted and retained inferred graph. A failure stops the release step. On success, a validation record binds the checked release to its ontology, shapes, Silver files, component RDF files, and serving graph. The publisher rejects missing or stale validation records.

Fingerprints normalize text line endings to LF so that Git checkout on Windows and Linux does not create false stale-release failures. This is a content-integrity check, not a cryptographic signature or a guarantee against someone deliberately editing both the inputs and validation record. Reproduction

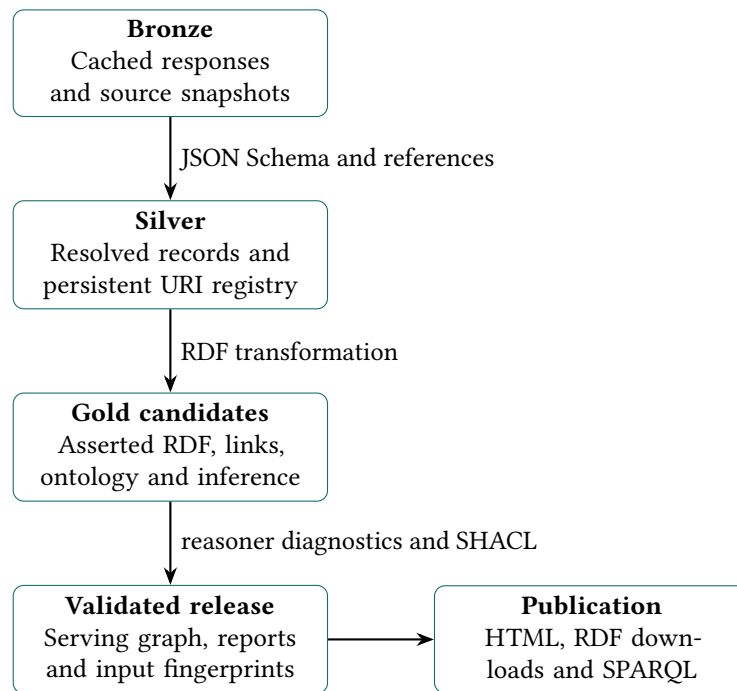


Figure 1: Release flow. A failed validation prevents replacement of validated inference and the serving graph; changed inputs invalidate the publication fingerprint.

of RDF is evaluated semantically; timestamps, blank-node labels, and serialization order can change without changing graph meaning.

4.2 Identity and URI persistence

A readable slug is used when a resource is first assigned a local URI. The revised minter consults the committed `uri_map.json` before allocating anything new. Renaming an entity or inserting another entity with the same label does not change a registered URI. Retired identifiers remain reserved. Source-key changes, such as a new QID replacing an article-based key, still require an explicit identity migration policy.

The previous person integration merged unidentified leaders by normalized personal name across all institutions. The audit found six names used across multiple institutions; splitting these records adds seven person records. This does not prove that seven new real people exist. It represents uncertainty conservatively: records without a shared identifier are separated by institution until identity can be established. The former ambiguous URI is reserved but is not asserted as identical to either replacement.

4.3 Provenance and disputed values

Institution records use `prov:wasDerivedFrom` to reference Wikidata items and specific Wikipedia revisions. Conflict and fallback logs provide additional evidence for some chosen fields. This is entity-level provenance plus partial field-level logging; the project does not yet attach complete provenance to every triple. A Wikidata entity URI is also not a pinned statement revision.

The earliest reported foundation year is the current integration policy and may refer to a predecessor. Five records contain a different year in the infobox foundation date. The revised exporter retains such a date under `vnedu:reportedFoundingDate`, rather than asserting it as an unqualified `schema:foundingDate`. This preserves evidence without pretending that the historical ambiguity has

been adjudicated. Distinguishing predecessor origin, legal establishment, and opening would require dated event records and authoritative supporting sources.

5 Transformation: from source records to RDF

Transformation is implemented in Python, not in Fuseki. The main pipeline calls `step3_integrate.py` to reconcile source records and `step3_transform.py` to construct RDF with `RDFLib`. It does not use RML, R2RML, an Apache Jena mapping engine, or a relational database. Figure 2 connects the operations to their actual artifacts.

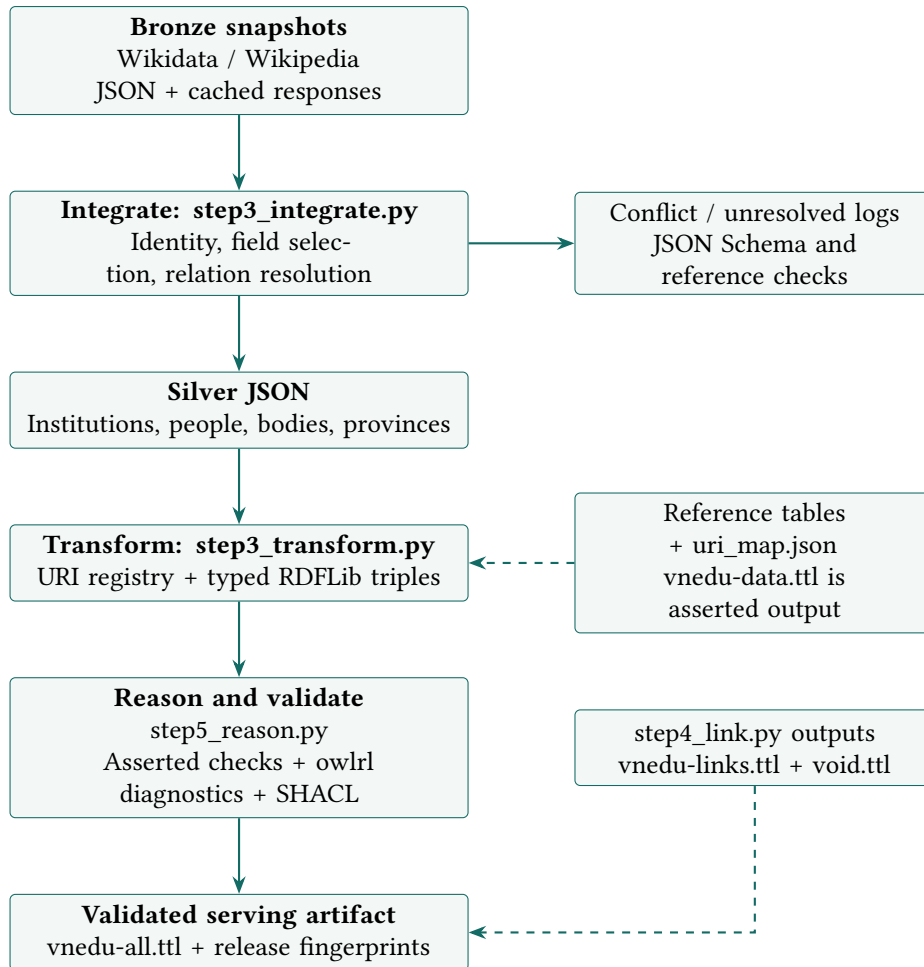


Figure 2: Implemented transformation and validation stages. Dashed arrows denote auxiliary inputs, not an independent source of factual verification. Fuseki is an optional consumer of the final graph, not a transformation step.

5.1 Integration policies and information loss

The integration script joins articles through Wikidata identifiers where available, normalizes names, parses dates and quantities, and resolves organization references. When a target is an educational institution, it becomes a membership or branch relation; a recognized public body becomes a governance target; and a company becomes an ownership target. Unknown targets go to an unresolved-reference report. A ministry associated with a record classified as private is moved to `stateManagedBy`. These are executable heuristics, not legal determinations. The original source predicate and temporal scope are not preserved on every resolved edge.

The earliest candidate founding year is selected and disagreements are logged. Integer counts lose the measurement context when no corresponding year or unit is retained. Some campus and funding values stay as source strings because their units are not reliably normalized. The source collector’s loss of birth-date precision cannot be repaired simply by assigning an RDF datatype later. Bronze, Silver and conflict reports must therefore be retained together when reviewing a disputed result.

5.2 Concrete mapping example

Table 4 follows the shipped Silver record Q3075696, Hanoi University of Science and Technology. The table demonstrates transformation, not an independent verification of the institution’s current legal status or leadership. Let *u*: abbreviate the project’s *resource/university/* namespace.

Table 4: Selected Silver-to-RDF mappings for a real input record.

Silver input	RDF output	Rule or caveat
key = Q3075696	Subject <i>u</i> :dai-hoc-bach-khoa-ha-noi	Reuse the persistent registry; do not mint again on a label change.
kind = University	<i>rdf:type</i> <i>vnedu:University</i>	Classification came from integration policy, not from Fuseki.
name_vi	<i>rdfs:label</i> and <i>skos:prefLabel</i> , tagged @vi	Keep language tags, not an untyped display string.
founding_year = 1956	<i>vnedu:foundingYear</i> "1956" <i>^^xsd:integer</i>	May refer to a predecessor’s origin; integer is used to remain inside OWL 2 RL.
province = Q1858	<i>vnedu:locatedIn</i> the registered Hanoi URI	An object IRI, not a province-name literal.
ownership = public	<i>vnedu:ownership</i> <i>vnedu:PublicOwnership</i>	Later classifies the subject as a public institution.
viwiki_revid = 75604914	<i>prov:wasDerivedFrom</i> a Wikipedia oldid URL	Revision-specific entity provenance; not per-triple attribution.

The exporter also emits coordinates as decimals, counts as non-negative integers, and websites as IRIs. Missing optional values usually produce no triple; they are not encoded as an empty string, zero, or a claim of nonexistence. Reference programs generate a program URI from the registered institution slug and major code. Consequently, this model identifies one illustrative offering per institution–major pair; it cannot distinguish multiple curricula, degree levels, delivery modes or academic years for that pair without extending the identifier design.

5.3 Linking, reasoning and release boundaries

`step4_link.py` writes a separate link graph and dataset metadata. The reasoner reads the local ontology plus asserted data, computes a closure, and retains selected instance-level entailments. Raw asserted constraints are evaluated against the unmerged asserted graph, independently of equality effects. The current script invokes this check after closure computation, but before replacing the inferred and serving files; it is not an early pre-computation performance gate. Successful validation produces `validated-release.json`. Publication checks the recorded input hashes.

The serving file is the set union of ontology, assertions, retained inferences, links and metadata. Separate files remain available, but the union itself does not carry named-graph boundaries. Loading that file into Fuseki’s default graph does not restore those boundaries or create assertion-level provenance.

A versioned dataset with named graphs and an explicit default-union query policy would be a future design change, not an existing feature.

This is a staged batch workflow, not a transaction spanning every artifact. Failed validation preserves the previous inferred and serving files, while preceding stages may already have rewritten candidate assertions and links. Hash checks protect the static publisher; the revised Fuseki loader enforces that same release gate. An immutable staging directory followed by an atomic release-pointer switch would make this boundary stronger.

6 Ontology design and relationship semantics

6.1 Design method

The ontology was designed from competency questions rather than from the visual shape of the source tables. Candidate concepts were grouped into modules, each relation was assigned a domain and range only when the data semantics justified it, and stronger axioms were kept separate from data-quality rules. The resulting model is therefore a TBox: it describes classes, properties, signatures, hierarchies, alignments, and selected entailment patterns. The institutional records, people, programs, and locations are the ABox. SHACL shapes and raw-record checks validate the ABox without pretending that every optional field is complete.

This distinction also explains why the ontology diagram should not be read as a relational database schema. A relational diagram emphasizes tables and cardinalities; an OWL diagram emphasizes concepts, property semantics, and possible entailments under the open-world assumption. The project uses an ER/UML-like projection for readability, while the Turtle axioms, Protégé inspection, profile check, and reasoner output remain authoritative.

6.2 Modules

The ontology has 39 named local classes, 38 local object properties, and 27 local datatype properties in the audited release. Table 5 summarizes the main relationship choices. Figures 12–14 visualize selected implemented axioms, not a proposed replacement ontology. Classes, property signatures and individual-level inferences are distinguished explicitly.

Table 5: Main modules and intended distinctions.

Module	Main entities or properties	Interpretation
Organizations	Institution, governing body, company; memberOf, branchOf, governedBy, ownedBy	Membership, branch status, governance, and ownership are separate relations.
Oversight	stateManagedBy	General state oversight is distinct from direct governance.
People	Person, InstitutionLeader, Alumnus; rector, director, councilChair	Leadership roles differ; education records do not certify graduation.
Geography	Province, FormerProvince, Region; locatedIn, partOf, mergedInto	Geographic crosswalks support aggregation under successor boundaries.
Education	AcademicProgram, Major, FieldOfStudy; offersProgram, ofMajor, inField	Example programs connect institutions to subjects; these are not an official offering inventory.
Identity	owl:sameAs, skos:closeMatch, foaf:isPrimaryTopicOf	Identity, approximate concept mapping, and document-topic links are not interchangeable.

Disjointness axioms separate selected organizational and geographic categories. Their usefulness depends on defensible category definitions. A statement that violates a disjointness axiom could indicate erroneous data or an overly rigid ontology, so a detected conflict should trigger inspection rather than automatic deletion of evidence.

Vocabulary alignments use subclass and subproperty relationships. Such alignments still carry logical implications; choosing a weaker alignment than equivalence does not make it semantically harmless. The current inference run does not import and validate the complete external ontologies. Its absence of detected conflicts therefore applies to the local reasoning scope, not every possible external combination.

6.3 Visual specification of ontology 2.2

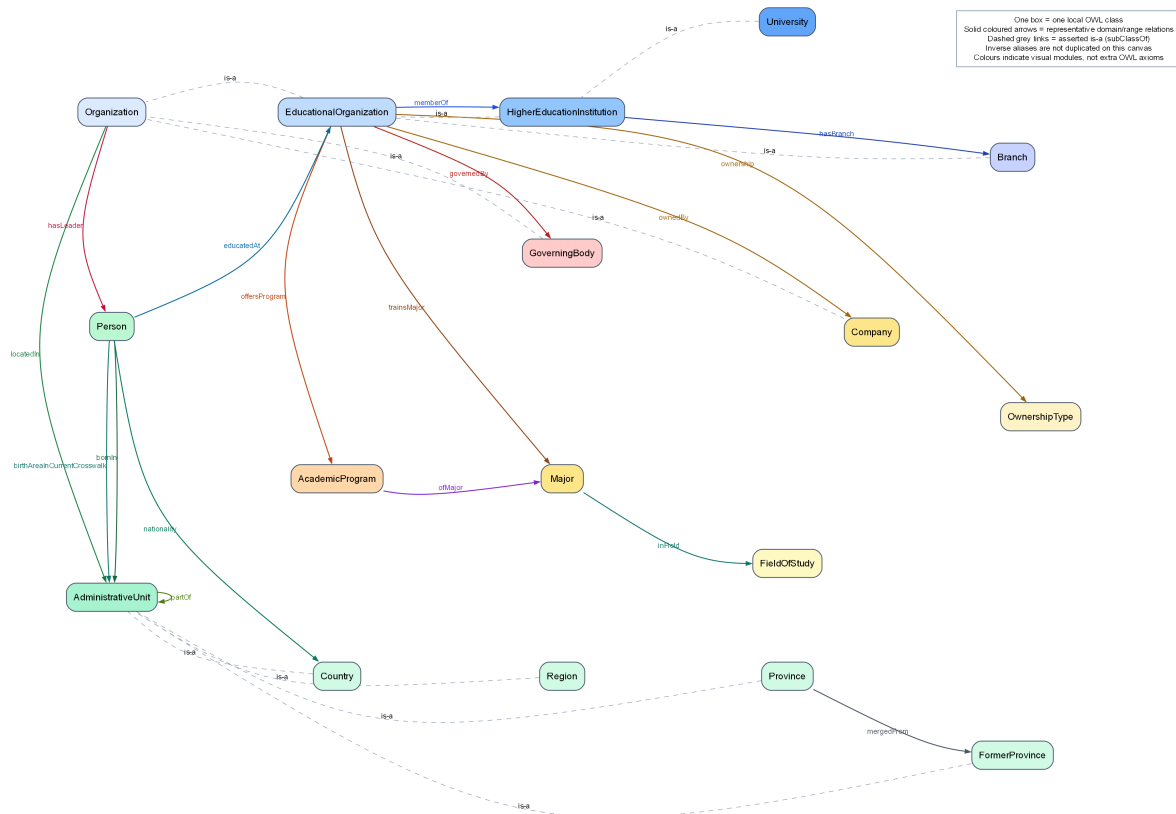


Figure 3: Main unique-node ontology map generated from the validated local OWL model. Each local class is rendered exactly once. Solid coloured arrows are asserted domain–range relations; dashed grey links are asserted `rdfs:subClassOf` axioms. The map uses one representative direction for inverse pairs so that it remains readable; the exact property inventory is documented separately. Coloured regions are presentation modules only and do not add OWL axioms.

The map is the primary human-readable description of the ontology. It is intentionally closer to an ER/UML or GraphRAG schema than to a raw triple dump: organizations, people, education, governance, ownership and geography are shown as distinct conceptual modules, while every class remains a single node. Its layout is generated deterministically from the Turtle source, not drawn by hand or inferred from a screenshot. The source model is checked independently with the OWL API bundled with Protégé, OWL 2 RL materialization, SHACL validation and SPARQL regression queries. Therefore the figure is a readable projection of checked axioms, not a substitute for the formal validation artifacts.

Protégé is retained as the direct ontology-inspection tool for class and property axioms. GraphDB and Apache Jena Fuseki are retained as execution checks for repository loading, reasoning and SPARQL end-

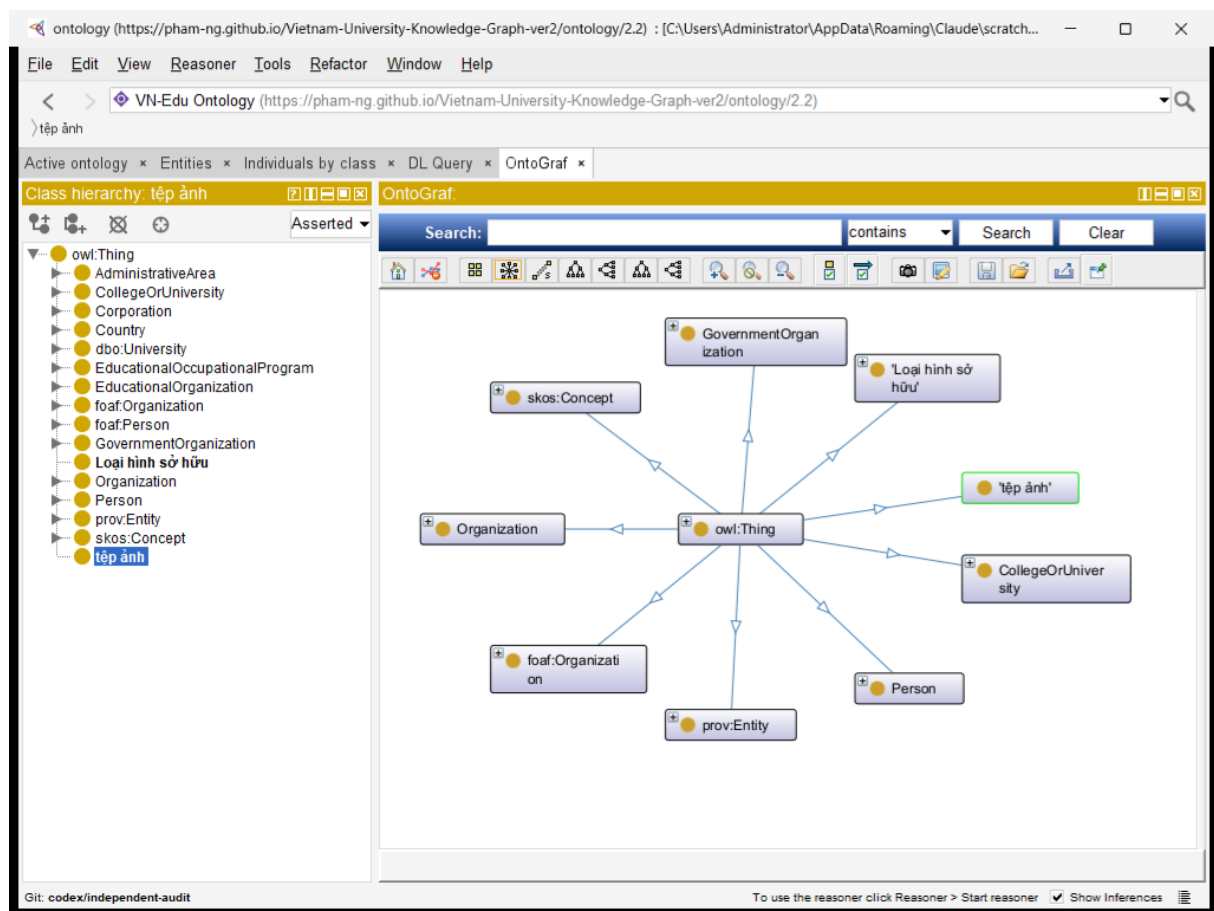


Figure 4: Direct screenshot of Protégé 5.6.7 with the OntoGraf plug-in, after loading the committed ontology/vnedu.ttl. The class hierarchy is shown at left and the selected ontology graph at right; the graph was laid out by OntoGraf and includes the selected OWL classes and their asserted subclass relations.

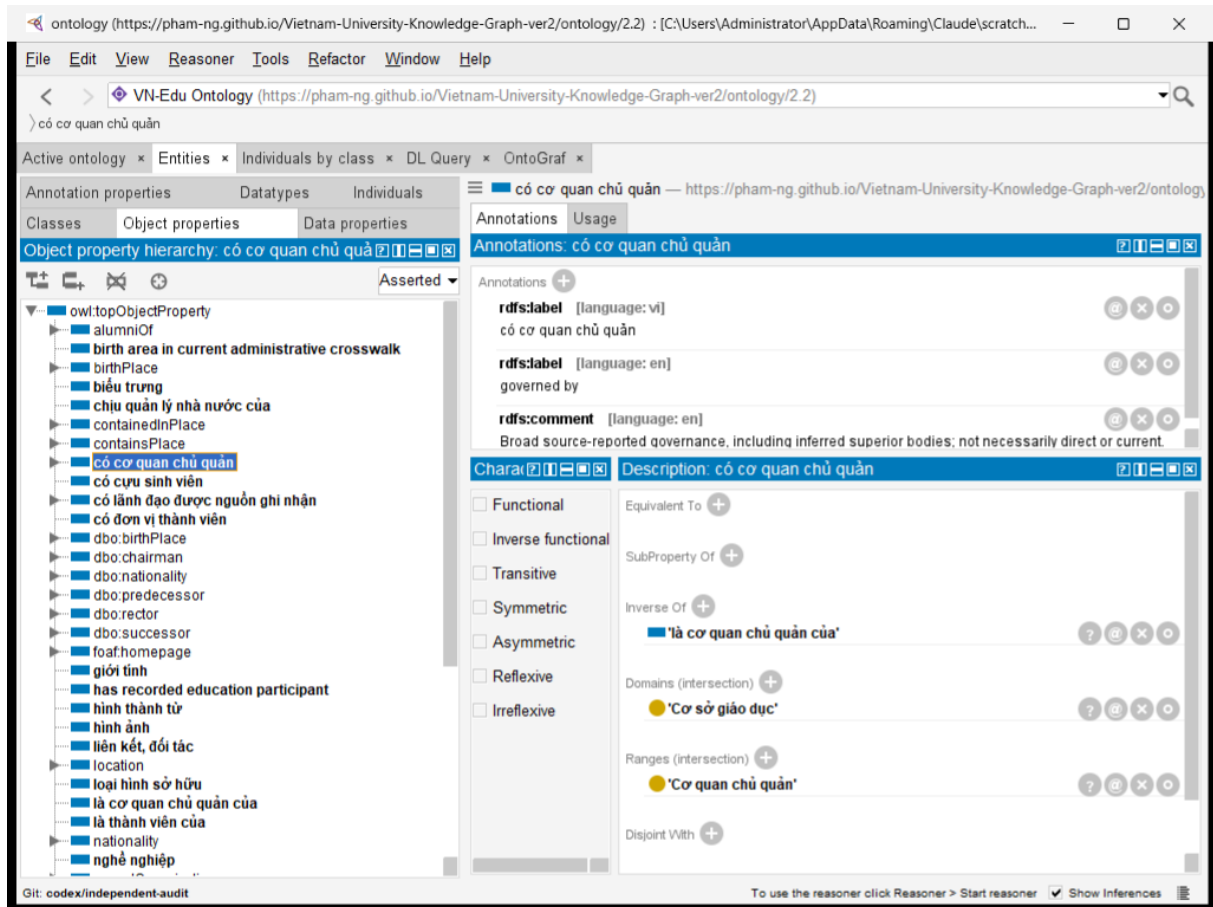


Figure 5: Direct screenshot of Protégé’s Object properties view for the committed ontology. The selected `governedBy` relation is displayed with bilingual labels, its comment, domain `EducationalOrganization`, range `GoverningBody`, and its inverse relation. This is the property-signature view used to verify that the graph has database-like domain–relationship–range discipline; it is not a hand-drawn diagram.

to-end behaviour; their screenshots are intentionally omitted from this section so that the main ontology explanation is not dominated by tool-specific layouts. This separation is important: a workbench layout is a user-interface projection, whereas the OWL axioms, reasoner output and query tests are the evidence for correctness.

The hierarchy and the relation map answer different questions. A root class is not an OWL error under the open-world assumption: it is a class for which this ontology has not asserted a local superclass. In particular, `Person`, `AdministrativeUnit`, `AcademicProgram`, `FieldOfStudy`, `OwnershipType` and `SourceObservation` are independent modelling concepts. The defined classes `PublicInstitution`, `PrivateInstitution`, `MilitaryInstitution` and `PoliceInstitution` are classified by `owl:equivalentClass` restrictions rather than by an asserted `rdfs:subClassOf` edge; adding a superclass only to make a picture connected would change the semantics. If a future competency-questions review confirms that those categories are intended as direct kinds of educational organization, that axiom can be added with a test case; it is not inferred from the visualization requirement.

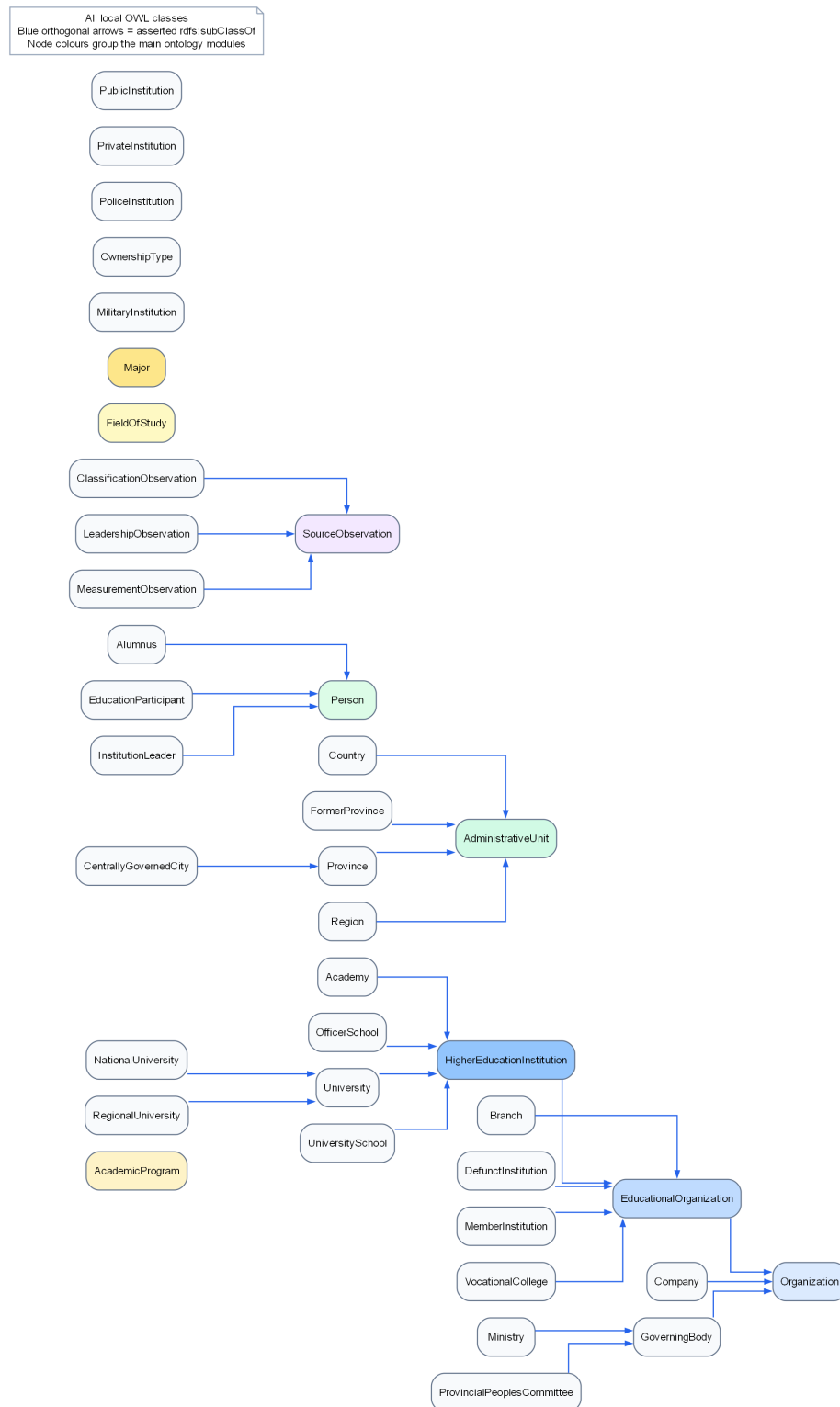


Figure 6: Complete local OWL class hierarchy generated directly from `ontology/vnedu.ttl`. Every displayed blue arrow is an asserted `rdfs:subClassOf` triple; disconnected nodes are intentionally retained because the ontology does not assert a superclass for them. Orthogonal routing is used to avoid crossing edges.

Each row is one asserted local object-property signature.
Left box = domain; diamond = predicate; right box = range.
An unspecified endpoint is shown explicitly, not invented.

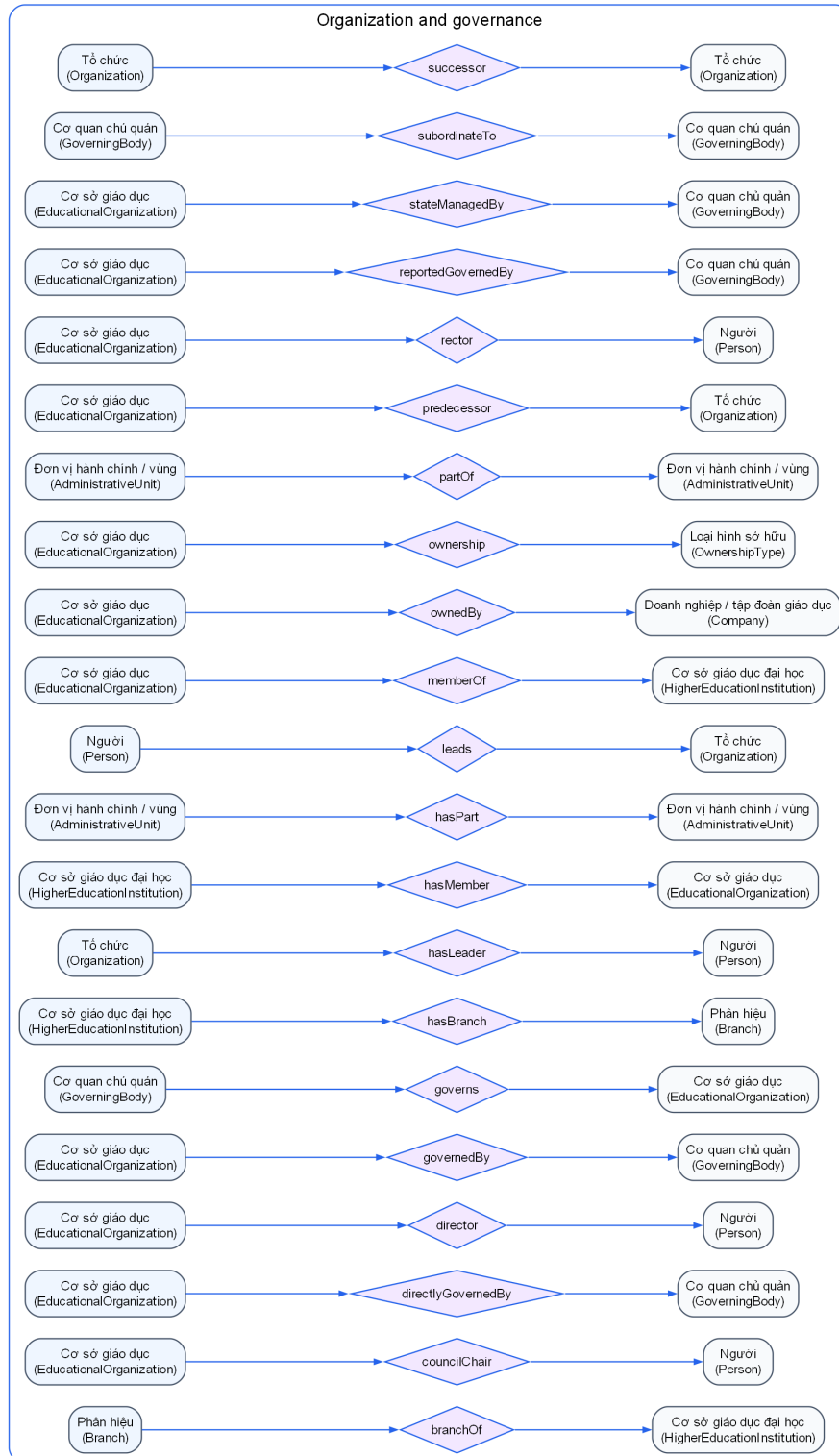


Figure 7: Exact local object-property signatures for organization, governance, membership, ownership and structural relations. Each row is independent: left box is the declared domain, the diamond is the predicate, and right box is the declared range. No edge is routed through another row.

Each row is one asserted local object-property signature.
Left box = domain; diamond = predicate; right box = range.
An unspecified endpoint is shown explicitly, not invented.

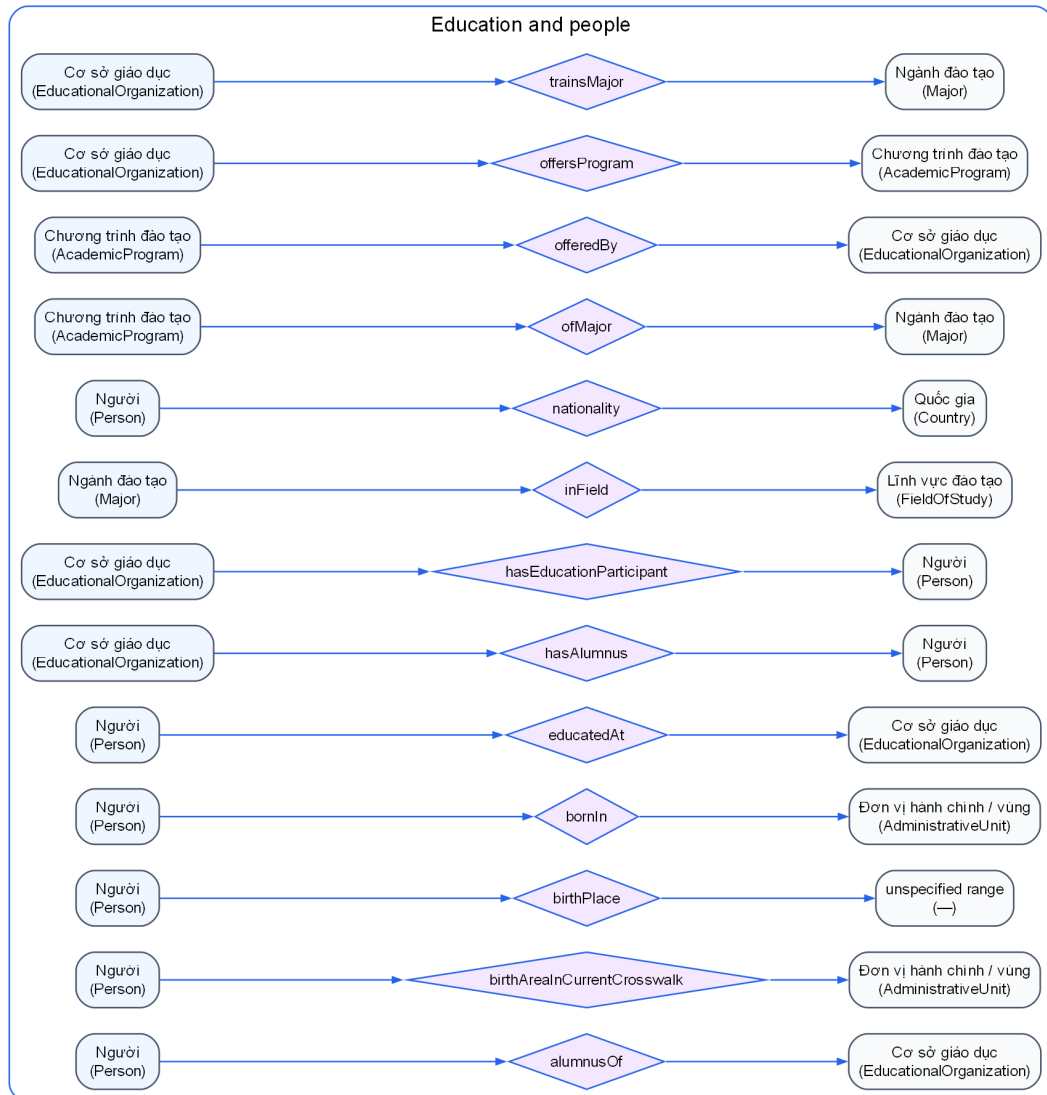


Figure 8: Exact local object-property signatures for programmes, majors, fields, people and educational participation. The missing range of birthPlace is shown explicitly as unspecified rather than guessed.

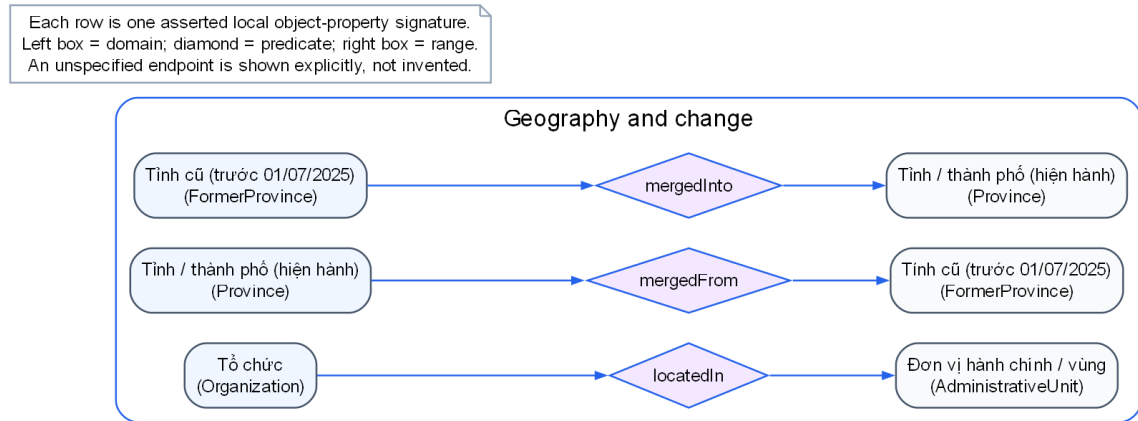


Figure 9: Exact local signatures for location and province-change relations. The former/current province distinction is represented as a directed relation, not as a timeless equivalence.

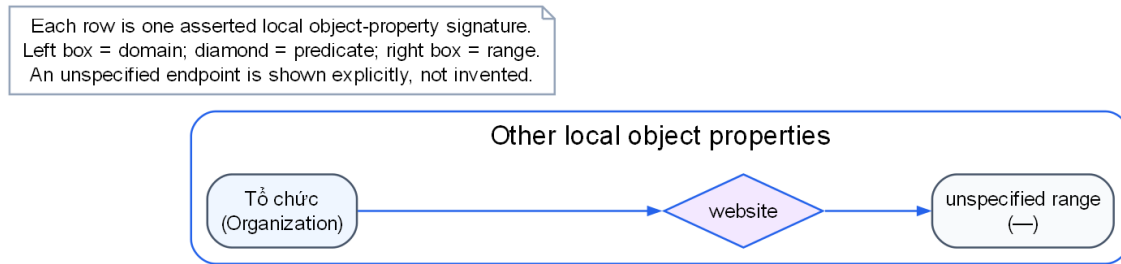


Figure 10: The remaining local object property, `website`, has an asserted domain but no asserted range in the ontology. This is shown as an open endpoint, not silently assigned `xsd:anyURI`.

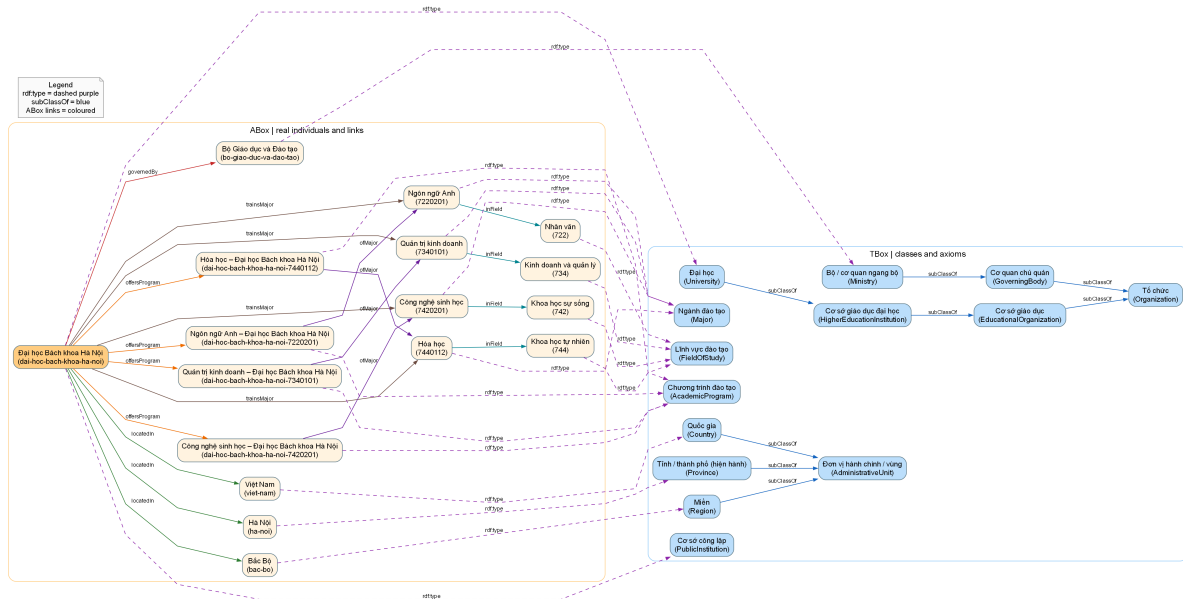


Figure 11: Data-driven ABox view generated from the gold serving union. The orange nodes are named individuals: one university, four programmes, four majors, four fields, location units and the reported governing body. Coloured edges are asserted or materialized object-property links; dashed `rdfs:type` edges connect only the most specific visible types to the TBox. This reproducible instance view complements the main unique-node TBox map and the direct Protégé inspection.

6.4 TBox, ABox and constraint semantics

The ontology is intentionally split into a schema layer and a data layer. The TBox is ontology/vnedu.ttl: classes, subclass axioms, property signatures, inverse properties, property chains, defined classes and selected disjointness. The ABox is the generated RDF in data/gold/vnedu-data.ttl, with data/gold/vnedu-all.ttl as the serving union of asserted data and materialized entailments. A URI such as university:dai-hoc-bach-khoa-ha-noi is an individual; vnedu:University and vnedu:HigherEducationInstitution are classes. Mixing these levels without explicit rdf:type and rdfs:subClassOf labels is what made the earlier graph misleading.

Table 6: Separation of ontology axioms, data assertions and validation constraints.

Layer	What is asserted	Correct interpretation
TBox	rdfs:subClassOf, rdfs:domain, rdfs:range, inverses, chains, restrictions and disjointness.	Schema-level meaning and inference rules; it does not require every possible individual to have every field.
ABox	Individuals, rdf:type, object-property links, literals, external identifiers and provenance observations.	Source-backed claims; an omitted triple is unknown under the open-world assumption.
OWL reasoning	Domain/range and property axioms materialize types and consequences, including inferred superclasses and inverse links.	A wrong domain/range can infer a wrong type; OWL entailment is not a database foreign-key rejection.
SHACL	shapes/vnedu-shapes.ttl checks cardinality, datatypes, patterns, temporal intervals and observation structure.	Closed-world data-quality validation, separate from OWL entailments.

The design therefore uses domain and range as semantic typing constraints, and SHACL for operational requirements such as “at least one provider” or a valid year range. This is the standards-correct combination for an RDF/OWL knowledge graph: imposing every relational-database NOT NULL or foreign-key rule directly in OWL would be unsound under the open-world assumption. Time-dependent legal categories remain qualified observations rather than permanent global disjointness axioms.

6.5 Implemented corrections and evidence limits

The design is standards-based and suitable for an educational knowledge-graph prototype, but consistency is not a certificate of factual accuracy. Version 2.2 makes the following changes without reminting existing entity identifiers.

Table 7: Ontology 2.2 corrections and their limits.

Concern	Implemented correction	Evidence boundary
Supervision versus inclusion	Removed parent-organization alignments from governance, subordination, ownership and membership. The branch alignment remains because a branch is a structural subdivision.	Supervision or investment does not automatically establish organizational parenthood.

Concern	Implemented correction	Evidence boundary
Direct versus indirect governance	Added <code>directlyGovernedBy</code> and <code>reportedGovernedBy</code> under the broad <code>governedBy</code> ; only the broad relation propagates through superiors.	Legacy source relations migrate to reported, not unsupported direct, governance.
Leadership versus employment	Removed leader-to-employee and leader-to-worksFor alignments; added role observations.	A council chair need not be an employee.
Institutional type changes	Removed global disjointness between the four legal-type classes; added classification observations.	The class projection is not a dated authoritative legal register.
Education versus alumni	Recorded education becomes <code>educatedAt</code> , with the neutral inferred class <code>EducationParticipant</code> .	Attendance does not entail graduation. Alumni terms remain for explicit evidence.
Joint programs	Removed functionality from <code>offeredBy</code> and <code>ofMajor</code> ; SHACL requires at least one provider and major.	Curated program identifiers still do not distinguish all curriculum variants.
Birth geography	Successor and containment chains produce <code>birthAreaInCurrentCrosswalk</code> , not additional <code>bornIn</code> facts.	A current province is not asserted to have existed at birth.
Version publication	Archived 2.1 and 2.2 Turtle; generated HTML, Turtle and JSON-LD version documents and dynamic routes.	The source and Pages workflow provide the route; stable public uptime still depends on deployment.

6.6 Standards and conformance assessment

The audited ontology follows the RDF 1.1/Turtle and OWL 2 RDF serialisation conventions and uses explicit OWL classes, object properties, datatype properties, inverse properties, selected functional properties, disjointness, property chains and defined classes. The current source contains 39 local classes, 38 local object properties, 27 local datatype properties and 862 ontology triples. The visualisation above is generated from those current source counts, so it can be regenerated after an ontology change rather than becoming a stale hand-drawn figure.

The design is suitable for a standards-based academic prototype, but it is not an automatic “five-star” certification. The following claims are deliberately separated:

Table 8: Standards assessment and remaining boundaries.

Area	Evidence in this release	Honest qualification
RDF and URI design	Stable HTTPS namespace, version IRI, Turtle/JSON-LD/N-Triples publication, content negotiation checks and licence metadata.	Deployment and long-term availability at the canonical origin remain operational responsibilities.
OWL 2 modelling	The ontology parses as RDF and passes the OWL API <code>OWL2RLProfile</code> gate with zero violations; inverse relations, chains, restrictions, equivalent classes and selected disjointness remain explicit.	This certifies profile membership of the local ontology, not complete imported-vocabulary semantics or factual correctness.

Area	Evidence in this release	Honest qualification
Constraint validation	SHACL shapes validate required fields, datatypes, temporal intervals and observation structure; the current suite reports zero violations under its policy.	Warnings reflect incomplete source coverage; SHACL validation is not proof of factual truth.
Provenance and time	PROV-O-style source links and reified evidence observations preserve snapshot context and optional temporal qualifiers.	The current Silver snapshot still lacks claim-level dates for many facts; an undated observation is not a current fact.
Vocabulary reuse	Conservative links to Schema.org, FOAF, DBpedia, SKOS, PROV-O and external identifiers.	External ontologies are not fully imported and reasoned over; alignments must remain semantically narrow.
Linked Data stars	Machine-readable publication, stable links, external links, licence and open formats are implemented.	The five-star scheme is a publication maturity heuristic, not a formal conformance test, and does not certify accuracy, security or production performance.

6.7 Qualified observations and temporal interpretation

The transformer creates 3,153 `SourceObservation` records, including leadership, measurement and classification specializations. Each reifies a triple through `rdf:subject`, `rdf:predicate` and `rdf:object`. Its SHA-256-derived URI includes the assertion, snapshot digest, source context and qualifiers. Identical evidence preserves the observation URI; changed evidence creates a different record. Entity URIs remain stable.

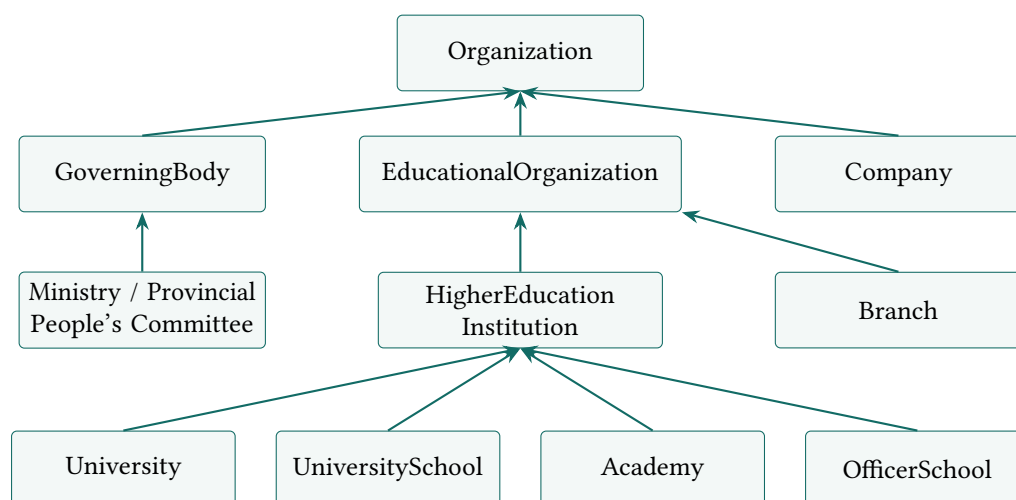
The implementation supports `validFrom`, `validThrough`, `referenceYear` and measurement units, validates calendar dates and interval order, and retains normalized SHA-256 digests of Silver inputs in `data/reports/observation-provenance.json`. Copied source references provide *entity-level context*, not proof that every assertion occurs in every referenced source. The snapshot digest is not an upstream publication date.

The present Silver snapshot does not retain sufficient temporal qualifiers for these measurements and appointments. Generated observations explicitly have unknown temporal validity. The system does not infer today's leadership or a current enrollment total; the institution infobox displays this limitation. Flat triples remain a source-snapshot projection. Historical queries must require relevant date bounds on observations; absence of an end date does not prove an appointment continues.

SHACL checks observation structure, allowed status, digest syntax, date datatypes and interval ordering. Regression tests also verify that reification alone does not assert the reified triple. This is a conservative migration, not a complete longitudinal collection system. Acquiring actual dates, claim-level provenance, foundation events and authoritative historical classifications remains future work.

6.8 Competency questions and prohibited conclusions

1. A direct body followed by its superior entails broad governance, not a second direct edge.
2. A council chair does not entail employment; an owner does not entail parenthood.



PublicInstitution and PrivateInstitution are defined by ownership values;
they are orthogonal to the four legal-type classes, not their replacements.

Figure 12: Selected organizational hierarchy. Arrows point from subclass to superclass. The four legal-type classes are no longer globally disjoint: an enduring institution can change type. This is a selected view of 39 classes.

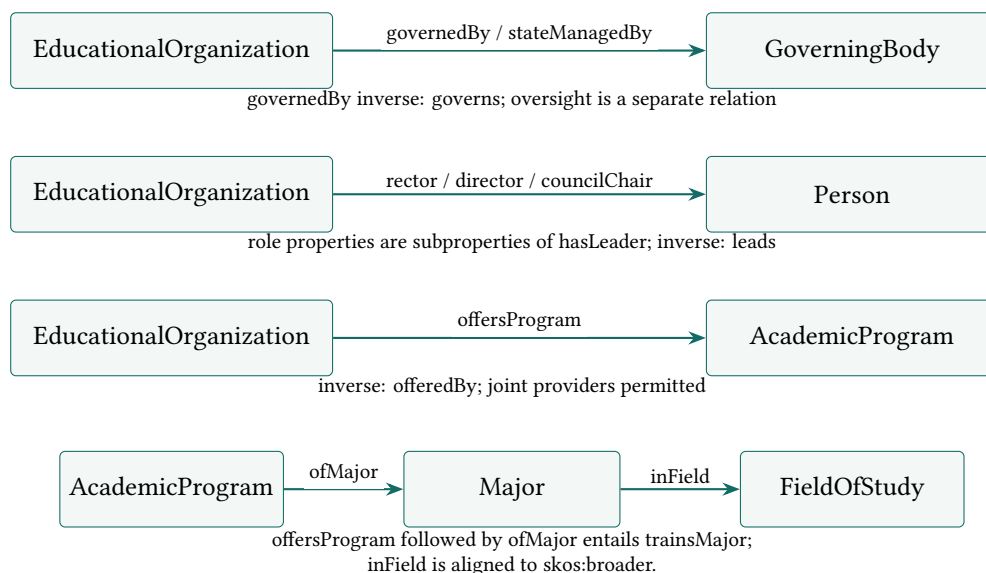
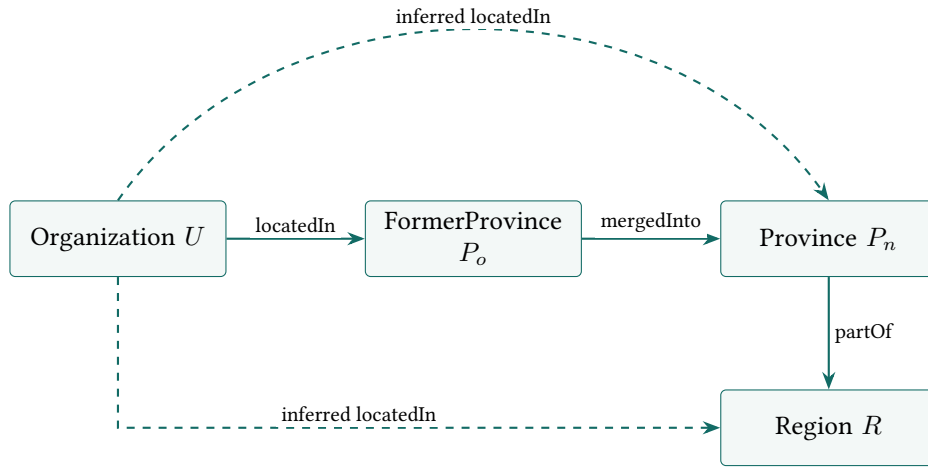
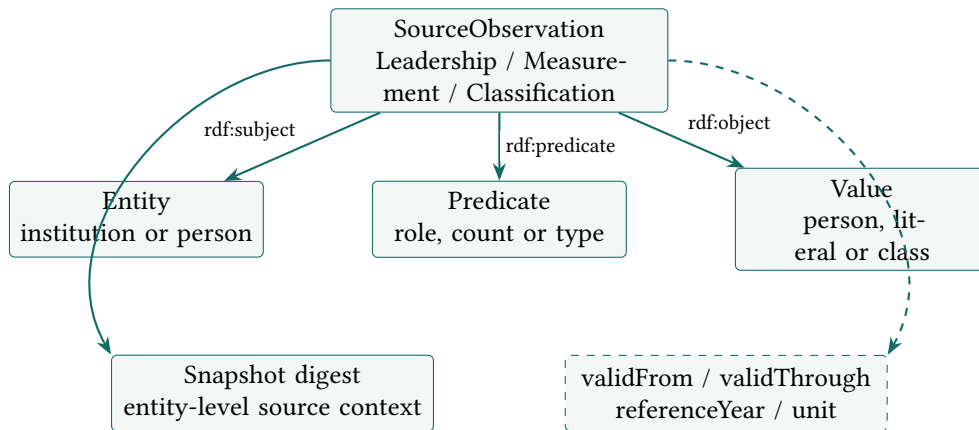


Figure 13: Selected domain-property-range signatures. These arrows are not mandatory foreign keys. Program providers and majors are no longer functional: multiple values must not force different institutions or subjects to become identical.



A current-boundary crosswalk is not a dated history.
The analogous bornIn chain does not prove that P_n existed at birth.

Figure 14: Institution-location inference under the selected administrative crosswalk. It does not establish historical validity. Birth-area mapping now uses a separate `birthAreaInCurrentCrosswalk` predicate instead of adding historical `bornIn` statements.



No supplied dates $\Rightarrow$ unknown validity, not current validity.

Figure 15: Observation model. Dashed fields are optional and must come from evidence. Missing dates mean unknown temporal validity. The diagram is schematic; no dates are fabricated for real officeholders.

3. Joint providers or majors do not become identical through a program.
4. Recorded education does not entail the alumni class.
5. A current geographic crosswalk does not create a historical birthplace assertion.
6. An undated observation does not answer a date-valid question merely because it is in the latest export.

Rule-engine execution, formal OWL profile membership, domain correctness and empirical coverage remain distinct claims. The formal OWL 2 RL gate is now asserted only for the committed release ontology, with its reproducible command and zero-violation evidence recorded in `data/reports/owl2rl-profile.txt`.

6.9 Materialization

The system computes a rule closure over the local ontology and asserted data and retains selected new instance-level statements. External identity links are joined into the serving graph after this step. This avoids equality-driven copying to remote identifiers, but means the published union is not a complete closure including all `owl:sameAs` effects. The reported inferred-triple count describes the retained subset, not the full internal closure.

Representative rules are:

$$\begin{aligned}
 \textit{locatedIn} \circ \textit{partOf} &\sqsubseteq \textit{locatedIn}, \\
 \textit{locatedIn} \circ \textit{mergedInto} &\sqsubseteq \textit{locatedIn}, \\
 \textit{governedBy} \circ \textit{subordinateTo} &\sqsubseteq \textit{governedBy}, \\
 \textit{offersProgram} \circ \textit{ofMajor} &\sqsubseteq \textit{trainsMajor}.
 \end{aligned}$$

For example, an institution assigned to a former province can also be related to the successor province and its larger region. Inverse properties expose the same relationship in the opposite direction without manually storing both directions in the asserted graph.

The model uses the `owlrl` implementation of OWL 2 RL/RDF rules. A separate OWL API `OWL2RLProfile` gate now checks the committed Turtle ontology; release 2.2 produced zero violations. Calendar years are represented as `xsd:integer`, and calendar dates as `xsd:dateTime` at midnight when the source supplies date precision only. This certifies the ontology's syntactic profile membership, not factual correctness, reasoner completeness over unimported external ontologies, or production performance.

6.10 Geographic and temporal limitations

The `birthAreaInCurrentCrosswalk` chains are geographic containment crosswalks, separated from `bornIn` in version 2.2. Relating a birthplace to a province created in 2025 does not mean that province existed when the person was born. Similarly, historical institutions should not automatically be counted as currently operating just because their location can be mapped to a current province. Queries for current institutions should explicitly consider dissolution evidence and its incompleteness.

Version 2.2 adds qualified observations for roles, measurements and classifications, but the available source snapshot still lacks sufficient temporal qualifiers and historical geographic versions. Absence of an end date is not proof of a current appointment. Birth-date collection currently loses source precision, and suppressing January 1 dates is only a heuristic: it can suppress real birthdays and cannot establish the precision of all other dates. These source-quality limitations remain documented work, not conclusions resolved by successful reasoning.

7 Linking and publication

This stage turns the validated graph into a Linked Data artifact. It has three distinct responsibilities: interlink local resources with external datasets, expose useful Web representations, and provide a query/storage runtime. The responsibilities are deliberately not collapsed into one product: the Python transformer creates RDF, the publisher creates resource documents, Protégé inspects ontology axioms, GraphDB is useful for interactive exploration, and Fuseki provides the tested Apache Jena SPARQL/TDB2 path.

Table 9: Technology roles in the VN-Edu publication architecture.

Component	Main role	What it does not prove
Python/RDFlib pipeline	Integrates records, assigns URIs, emits RDF, materializes selected entailments	Does not provide a persistent public endpoint by itself
Protégé and OWL API	Inspects ontology axioms and checks OWL 2 RL profile membership	Does not validate all ABox facts or measure production capacity
GraphDB Workbench	Interactive repository loading, visualization, and SPARQL exploration	A layout is not a formal ontology specification
Apache Jena Fuseki/TDB2	HTTP SPARQL/Graph Store access and persistent acceptance path	Does not mint canonical IRIs or establish factual accuracy
Silk Framework	Candidate entity-linking experiment on a frozen sample	Candidate links are not automatically accepted as owl:sameAs

7.1 External relations

The link graph contains 3,558 triples. Wikidata, ROR, GeoNames, and selected DBpedia links use owl:sameAs; subject mappings use skos:closeMatch; Wikipedia article links use foaf:isPrimaryTopicOf. The last of these relates an entity to a document and must not be counted as an identity assertion. Repeated attempts to add the same link no longer inflate linkset counters.

Shared identifiers are useful identity evidence, but errors and historical scope mismatches can occur in the source mappings. Some organizational links are discovered through label matching with contextual filters. These deserve separate manual review. The audit does not certify every identity link or every administrative boundary match.

7.2 Representations and deployment

The static publisher generates HTML pages, embedded JSON-LD, and RDF files. The dynamic server provides query handling and RDF content negotiation. The canonical namespace currently belongs to GitHub Pages; the dynamic service has another origin. Content negotiation at the dynamic service does not establish that the canonical static URI performs the same negotiation.

The implementation supports practices associated with five-star Linked Open Data. Section 9 assesses each level and reports a small, dated HTTP sample; it does not reuse the earlier report's live availability measurements. The repository declares data licensing and retains media metadata; this audit is not a legal verification of the license of every upstream asset.

The publisher now checks its output path before recursive replacement. It refuses the project root, external paths, symbolic links, and existing directories without generated-site markers. This prevents a misconfigured output environment variable from deleting source data or the project itself.

8 Deployment and Apache Jena Fuseki

This section documents how the validated RDF release is published and queried. It is useful to separate the URI design from the runtime: a URI identifies a resource, while Fuseki stores and queries triples about that resource. The same graph can therefore be published as static RDF and loaded into Fuseki without changing its identifiers. Fuseki is a SPARQL server, not a URI minting service [9]. The transformer assigns canonical identifiers before data enters any store. The same university IRI remains in RDF whether queried by RDFLib, Oxigraph or Fuseki.

8.1 Identity, representation and storage

```
# RDF subject identifying a university:
https://pham-ng.github.io/Vietnam-University-Knowledge-Graph-ver2/resource/university/dai-hoc-
↳ bach-khoa-ha-noi
# Local query endpoint:
http://localhost:3030/vnedu/sparql
# Graph Store Protocol target:
http://localhost:3030/vnedu/data?default
```

Uploading RDF does not make its canonical IRIs dereferenceable automatically. The publisher independently creates HTML, Turtle and JSON-LD descriptions. Ontology terms retain hash IRIs; entity and observation identifiers use slash paths. Immutable 2.1 and 2.2 version artifacts are now generated under ontology/. Dynamic version routes support RDF content negotiation and return 404 for unknown versions.

GitHub Pages remains a separate HTTP origin from Flask. Its static representation links and embedded JSON-LD do not implement Flask’s negotiation, nor the entity-to-document 303 pattern discussed in *Cool URIs* [6]. Adding Fuseki cannot fix those origin-level behaviors. Version artifacts are implemented and tested locally; their production deployment requires promotion of the review branch.

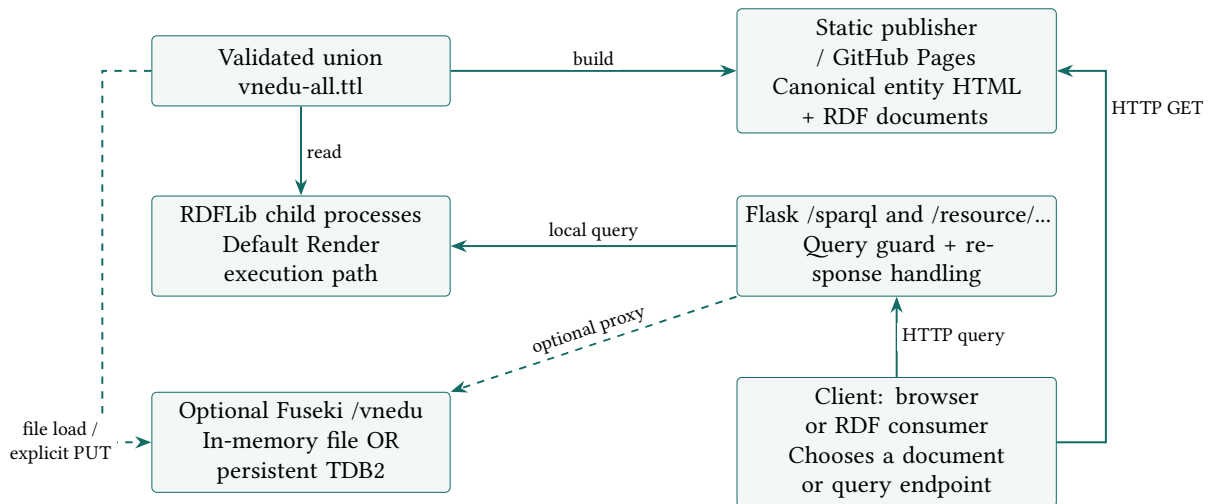


Figure 16: Publication and storage paths. The optional Fuseki branch has verified Linux and Windows acceptance runs; Render still explicitly selects local RDFLib. Storage does not replace the canonical identifiers.

8.2 Safe loading and reproducible runtime

The revised loader requires a successfully validated release before network writes. It targets an existing explicitly named dataset rather than creating one implicitly. An empty graph can be loaded directly.

Replacement of a nonempty default graph requires both `--replace` and a new `--backup` path; the backup is created exclusively, so an older backup cannot be silently overwritten. HTTP redirects and error responses are rejected. After PUT, the loader retrieves the entire default graph and checks RDF graph isomorphism, including blank nodes, rather than relying on a triple count.

Graph Store PUT replaces the selected graph [11]; named graphs are untouched by this loader. Credentials are read from environment variables, not command-line passwords. A failed comparison preserves the backup for operator recovery; it is not a distributed transaction or automatic rollback. Operators must stop concurrent writers during a release: a backup followed by PUT does not provide compare-and-swap concurrency control.

The runtime setup pins Apache Jena Fuseki 6.2.0 and verifies its published SHA-512 digest. The local launcher requires Java 21, validates the release, binds to loopback, and enables only query access with a configured timeout. Its file mode reloads the graph at startup; the separate acceptance harness explicitly tests persistent TDB2. The optional legacy Docker recipe now requires an operator-selected image and password, and binds port 3030 only on loopback. That Docker configuration was not the tested runtime.

8.3 Production-like security and performance acceptance

The additional `audit/production_readiness.py` test starts the same Waitress WSGI entry point used by `app/server.py --prod` with the read-only local backend. Under normal bounded concurrency (eight requests, two query workers), all eight health requests returned HTTP 200; the 8 October 2026 rerun measured 3,965 ms at p50, 3,996 ms at p95, and 3,996 ms maximum on the Windows review machine. An intentional eight-request overload returned two HTTP 200 responses and six HTTP 503 responses, demonstrating bounded back-pressure rather than unbounded queue growth. Security probes rejected a path-injection attempt (404), an unapproved SSRF SERVICE target (400), and an oversized query (413). The response headers now include `nosniff`, `SAMEORIGIN`, `strict referrer-policy`, `Permissions-Policy`, `CSP on HTML`, and an `X-Request-ID`; HSTS is enabled when TLS is detected or explicitly forced behind a trusted TLS terminator.

These numbers are a reproducible loopback acceptance sample, not a capacity SLO, penetration test, TLS/reverse-proxy audit, cloud IAM review, or production security certification. The process-per-query RDFLib backend deliberately trades throughput for isolation; a real deployment still needs external rate limiting, resource quotas, TLS termination, monitoring, authentication policy, and a load test against its selected Fuseki/WSGI topology. The full machine-readable evidence is `data/reports/production-readiness.json`.

8.4 Production hardening profile and release gates

The review branch adds a deployable reference profile in `deploy/` and records its scope in `docs/production-readiness.md`. The intended production path is:

Internet -> Caddy/Nginx TLS -> Waitress application -> private Fuseki -> encrypted TDB2 volume

The application layer is read-only for SPARQL, limits request bodies and serialized result size, applies an optional per-client rate limit, restricts CORS to an explicit allow-list, and emits request identifiers for incident correlation. The reference Compose file also uses a non-root application user, a read-only container filesystem, a private network, dropped Linux capabilities and a no-new-privileges flag. Fuseki image digests and admin credentials are deliberately operator-supplied; the public reverse proxy must never expose Fuseki administration or update endpoints.

These are implementation controls, not evidence that the stack has been deployed on a public production host. Before a production availability claim, the deployed Fuseki path must be tested for sustained and burst requests, p50/p95/p99 latency, CPU/memory/disk growth, query cancellation, rolling restart and backup restoration. The current local RDFLib p95 is therefore reported as an isolation/back-pressure observation, not as a capacity target.

Table 10: Production release gates that remain separate from OWL and SHACL conformance.

Gate	Required evidence before an authoritative production claim
Security deployment	Public TLS termination, trusted proxy configuration, firewall/IAM policy, secret rotation and external rate limits.
High-load performance	Deployed Fuseki benchmark with representative SELECT/CONSTRUCT queries, p50/p95/p99, saturation behavior and resource measurements.
Recovery	Backup restore, rolling restart and graph-isomorphism verification after replacement.
Factual accuracy	Independent adjudication of the five founding-year conflicts and seven unresolved values, plus a held-out identity-link evaluation.
Temporal validity	Claim-level provenance and validity intervals for leadership, enrolment and legal status; undated snapshots must not be presented as current.
Source authority	Official-register or institution-owner evidence for records currently supported only by encyclopedic or inferred sources.

The release can pass the OWL 2 RL profile gate, SHACL validation and regression tests while still failing these operational and factual gates. Keeping those gates separate is essential: OWL entailment is not a truth certificate, and a five-star Linked Data publication is not a five-star security or accuracy certification.

8.5 Executed end-to-end acceptance evidence

The primary reproducible test ran on Linux GitHub Actions with Fuseki 6.2.0 and Temurin Java 21.0.12.1, on 7 October 2026 at 01:50 UTC. A second run on the Windows desktop used the same Fuseki JAR, Java 21 and the Windows selector/socket workaround at 07:22 UTC; the launcher smoke test passed at 07:31 UTC. The tested code revision before these launcher edits was `ab0033d`; the current branch records the subsequent launcher fix. The input hash, JAR hash, runtime details and assertions are archived in `data/reports/fuseki-e2e-linux.json` and the Windows run in `data/reports/fuseki-e2e-windows.json`; the corresponding Linux CI run completed successfully.

The test uses a newly created disposable directory and ephemeral loopback port; no existing user dataset is overwritten. Zero dated-leadership rows is an honest reflection of missing date evidence, not a failed lookup to be filled with invented dates.

On Windows, the first attempt exposed a JDK 21 selector/socket initialization failure. The repaired launcher now sets the selector provider and short Unix-domain socket directory through inherited Java options (avoiding PowerShell 5.1 argument concatenation); the same disposable TDB2 acceptance test

Table 11: Measured Fuseki/TDB2 acceptance results.

Check	Observed result
Load complete validated serving graph	73,628 triples; graph-isomorphic to local RDF.
Attempt unapproved replacement	Refused before PUT.
Approved replacement with backup	Backup graph-isomorphic to previous contents.
Query parity with RDFLib	Count: 1 row; class totals: 68 rows; programs: 78 rows; explicitly dated leadership: 0 rows.
Stop and restart with the same TDB2 directory	Entire default graph remains isomorphic.
Restart without update permission	SPARQL Update and Graph Store PUT both return HTTP 405.
Read after rejected writes	Graph unchanged and isomorphic.

then passed all checks, and a direct query-only launcher smoke test returned HTTP 200. This proves local Windows operability, not production deployment. Configured timeout is not proof of cancellation under load. Authentication, SERVICE egress, concurrent writers, memory limits, disaster recovery and production TLS require separate acceptance tests. The verified read-only mode is access restriction, not a full security certification.

9 Five-star Linked Open Data assessment

The five-star scheme evaluates how data is published and connected, not the overall quality of a program or the correctness of every ontology axiom [7]. There is no Fuseki-specific requirement: RDFLib or static RDF publication can support the relevant mechanisms. Table 12 assesses the project’s data publication rather than assigning an unsupported software-quality score.

Table 12: Evidence against the cumulative five-star publication scheme.

Level	Criterion, summarized	Project evidence and qualification
1	Publicly accessible, openly licensed	Public sample accessible; dataset metadata declares CC BY-SA 4.0. Upstream media retain their own licenses.
2	Structured, machine-readable	JSON records and parsed RDF graphs, not only screen images or PDFs.
3	Non-proprietary format	JSON, Turtle and JSON-LD artifacts.
4	Web identifiers and open Semantic Web standards	Stable-registry HTTP IRIs, RDF descriptions and query interfaces; sampled entity documents parse. Canonical negotiation remains static; version artifacts are now implemented on the review branch.
5	Connections to other datasets	Corrected local link graph contains 2,309 owl:sameAs, 61 skos:closeMatch and 1,188 document-topic links. Existence is measured; every identity assertion is not independently verified.

Verdict. The implementation and sampled public RDF support the basic five-star publication pattern, subject to the declared licensing scope. This is not an independent certification, proof of factual accuracy, or proof that the corrected branch is deployed. Deployment, temporal-evidence and operations limits should be reported alongside the claim, not hidden by a five-star badge. Failing a single endpoint probe does not automatically reduce an otherwise accessible RDF dataset to a lower star level.

9.1 Dated HTTP evidence

audit/check_publication.py performs seven read-only requests with bounded timeouts; the result is saved in data/reports/publication-check.json. This historical, pre-migration sample was recorded

on 7 October 2026 at 00:10 ICT (UTC+7). The institution tested was `resource/university/dai-hoc-bach-khoa-ha-noi`; this is not a random or exhaustive sample.

Table 13: Observed HTTP behavior; deployed content may predate the corrected branch.

Request	Observation
Canonical entity, Accept HTML	HTTP 200, HTML, embedded JSON-LD and alternate RDF links.
Canonical entity, Accept Turtle	HTTP 200, still HTML; no HTTP redirect.
Entity's <code>.ttl</code> document	HTTP 200, Turtle, 204 triples; graph describes the entity IRI.
Entity's <code>.jsonld</code> document	HTTP 200, JSON-LD, 204 triples; graph describes the entity IRI.
Declared ontology/2.2 version	Source build emits immutable HTML, Turtle and JSON-LD; live Pages availability awaits deployment of this corrected branch.
Render <code>/healthz</code>	Read timeout after 15 seconds; health and deployed commit were not established.
Local Fuseki <code>/\$/ping</code>	Connection refused on port 3030; no reachable service at the configured address.

A Render timeout can reflect a cold start or a network/service problem; it is not proof of permanent downtime. The probe did not test SPARQL federation, persistence, admin credentials, or the full set of resource identifiers. The existing public deployment remains distinct from the audited branch, whose Linux CI and local tests validate a different scope.

The media distinction also matters. For example, the shipped HUST logo metadata records “fair use” while its campus photograph records CC0. Describing or linking such an image does not relicense the image under the dataset license. The revised publisher shows a source link, not an embedded image, for unknown, fair-use or unrecognized licenses. A data publication claim must not be expanded into a blanket claim that every remotely displayed asset is openly licensed.

10 Application, security and CI/CD

The repository treats the validated graph as a release artifact. The main CI workflow rebuilds the site, runs the reasoning and publication gates, and executes the regression suite. Separate workflow paths exercise Fuseki/TDB2 and the Silk runtime when their relevant inputs change. This split keeps the ordinary documentation build fast while preserving explicit evidence for the two external runtimes. A green workflow means that the declared checks passed on the CI runner; it is not a claim that a public deployment is secure or available.

The original SPARQL policy searched for textual patterns containing whitespace after `SERVICE` and `FROM`. Valid forms without that whitespace bypassed the filter. The audit reproduced this locally using loopback targets without issuing an outbound request. The revised policy inspects parsed SPARQL operations and permits only two exact HTTPS service endpoints. It rejects dataset-loading clauses, variable service targets, alternate ports, credentials, and arbitrary paths. Security keywords inside string literals are not treated as operations. The local outbound HTTP handler also rejects redirects.

The previous timeout waited only for `Graph.query` to return. `RDFLib` can evaluate `SELECT` bindings lazily, so result evaluation could continue beyond the limit. Cancelling a future also cannot stop an already running Python thread. The revised local backend evaluates and serializes each query in a disposable process, terminates it on timeout, and limits concurrent workers. An expensive Cartesian-product regression test checks that capacity is released and a subsequent small query succeeds.

The local backend limits serialized responses to 8 MiB. This does not impose a hard memory limit during intermediate evaluation, and parsing the public request still occurs in the web process. A hostile-workload production deployment therefore needs process or container memory limits, request-rate

Table 14: CI/CD gates and the claim supported by each gate.

Gate	Executed check	Supported claim
Reasoning and release metrics	OWL/RL materialization, consistency diagnostics, SHACL, graph union checks	The checked local release satisfies the implemented structural policy
Static publication	Site build, fingerprints, HTML/RDF/JSON-LD generation	The generated publication is reproducible from the validated inputs
Regression suite	Python tests for transformation, queries, security, URIs and release gates	The known counterexamples remain covered by automated tests
Fuseki/TDB2 workflow	Load, read, query parity, restart, backup, and denied update tests	Apache Jena persistence and read-only acceptance succeeded in the test environment
Silk workflow	Frozen candidates and reference-agreement metrics	Silk executed and produced candidates; independent identity accuracy remains unproven

controls, and appropriate upstream isolation. The process-per-query approach also reloads the graph and has a throughput cost. Earlier in-memory latency measurements do not describe this implementation.

The Fuseki proxy supplies an execution timeout parameter and does not follow proxy redirects. Actual query cancellation and remote SERVICE egress remain responsibilities of the configured Fuseki instance. The separate Linux acceptance run verifies load/query/persistence/read-only behavior; it does not verify production federation controls. Raw POST requests with `application/sparql-query` are now accepted, alongside form-encoded query requests.

11 Evaluation and results

11.1 Release measurements

All values in Table 15 are generated by `audit/release_metrics.py` from the validated local release. That script also checks that the serving graph is isomorphic to the union of its declared components. It does not request data from a remote endpoint.

In the corrected snapshot, selected SHACL checks report 0 violations, 34 warnings, and 1 informational result. Warnings include missing foundation years and institution locations. Warning-tolerant conformance means the configured release policy permits these omissions; it does not mean all records are complete. The consistency diagnostics report no detected local violations after the fixes.

Field coverage uses all 300 institution records as its denominator: 278 have a selected foundation year, 288 a province reference, 256 an ownership value, 237 a website, 175 coordinates, 60 an admission code, and 28 a student-count value. These are presence measurements, not estimates of truth, timeliness, or national coverage. In particular, 271 higher-education classifications and 300 educational-organization records are different denominators and should not be interchanged.

11.2 Link experiment: agreement, not independent accuracy

The available experiment uses 86 institutions with English labels and identifier-derived DBpedia reference targets. A local token-Jaccard matcher selects the highest-scoring target from that closed reference candidate set. Table 16 reports agreement with the reference frozen from revision 58c65f0. This baseline is not Silk; the separate actual Silk execution is reported below.

Table 15: Measured size of the corrected local release.

Quantity	Count
Educational-organization records	300
Classified higher-education institutions	271
Person records	1,702
Institution-leader records	221
People with recorded education relations	1,481
Current / former province-level units	34 / 29
Illustrative programs / majors	78 / 37
Asserted data triples	47,291
Retained inferred triples	21,825
Link-graph triples	3,558
Ontology / dataset-metadata triples	862 / 96
Distinct triples in serving union	73,628
Local resource IRIs appearing as subjects	5,401

Table 16: Token-Jaccard agreement with the identifier-derived reference.

Threshold	Predictions	Disagreements	Ref. precision	Ref. recall	Ref. F1
0.2	85	13	84.7%	83.7%	0.842
0.4	80	8	90.0%	83.7%	0.867
0.6	76	6	92.1%	81.4%	0.864
0.8	68	4	94.1%	74.4%	0.831
1.0	58	1	98.3%	66.3%	0.792

Let M be the set of predictions, R the reference pairs, and $TP = |M \cap R|$. The reported precision is $TP/|M|$, recall is $TP/|R|$, and F1 is their harmonic mean. Calling R independent ground truth would be incorrect because the identifier linker generated it. Scoring that same linker against R yields perfect agreement by construction. The previous claim of 100% linking accuracy is therefore withdrawn.

The selected sample excludes unlinked institutions and targets outside the reference set. These restrictions affect task difficulty and coverage. The threshold with the highest F1 on this sample is a descriptive observation, not a validated threshold chosen on a separate development set. A defensible accuracy study needs independently annotated links and non-links, dated evidence, an adjudication procedure, and a held-out test set.

11.3 Executed Silk Framework experiment

A separate experiment now executes actual Silk Framework 3.6.0, rather than referring to the Python Jaccard baseline as Silk. The official single-machine entry point was compiled with Scala 2.12.10 against released workbench libraries and run on Java 8. Pinned checksums are in `audit/setup_runtimes.ps1`; the runner, XML rules, input N-Triples, candidate outputs and result JSON are included in the repository. The reproducible execution is scheduled in `.github/workflows/silk-e2e.yml` on Ubuntu with Java 8. On the reviewed Windows desktop, Java 8.351 and 8.481 both timed out before producing candidates; this is recorded as an environment limitation, not converted into a success claim.

The experiment freezes the 86-source/86-target sample from revision 58c65f0, giving 7,396 possible pairs. Both methods use the same lowercased English labels. Silk uses edit-distance thresholds of 0, 3 and 8 with a top-one filter, while the Python baseline uses token-Jaccard similarity. Their threshold values are not numerically comparable. The XML filter is inside `LinkageRule`, as required by the 3.6.0

parser; assertions verify at most one prediction per source and the experiment-only predicate. Older example configurations are not assumed equivalent.

Table 17: Actual Silk versus the fixed-sample Python baseline: reference agreement only.

Rule	Predicted	Matches	Ref. precision	Ref. recall	Ref. F1
Silk: distance 0	57	57	100.0%	66.3%	0.797
Silk: distance 3	64	63	98.4%	73.3%	0.840
Silk: distance 8	75	70	93.3%	81.4%	0.870
Jaccard: similarity 0.4	80	72	90.0%	83.7%	0.867
Jaccard: similarity 0.6	76	70	92.1%	81.4%	0.864
Jaccard: similarity 0.8	68	64	94.1%	74.4%	0.831

These results do not estimate independent identity accuracy. Some historical reference identities are rejected by the current stricter linker; they remain in the frozen experimental reference, not in accepted release identities. The sample excludes open-world negatives, lacks independent adjudication and has no held-out tuning set. Silk’s tie handling is engine-defined; Python uses descending target URI for ties. Small differences do not establish general superiority.

No prediction is promoted to `owl:sameAs`. Silk emits a candidate-only URN predicate and its files are outside the serving pipeline. It is a research extension, not a normal publication dependency. Neo4j is not added: a second graph model and synchronization path would supply neither missing time evidence nor another five-star requirement. The implementation follows the official Silk source and link-specification documentation [8].

11.4 Regression evidence

The audit adds counterexamples for compact SPARQL syntax, exact service endpoint policy, keywords inside literals, raw query POSTs, cancellation and worker recovery, datatype diagnostics, literal ranges, publication gates, stale release records, persistent URIs, same-name identity collisions, offline cache misses, impossible dates, and dangerous publisher paths. Eight initial counterexamples failed before correction and passed after the relevant fixes. The final test-run result and command are recorded in the companion audit notes rather than a manually maintained badge.

The full pipeline is reconstructed from the shipped cache, and both static-site variants are built before the server integration suite. A successful run establishes that the tested release follows the intended local contracts. It does not constitute a penetration test, an independent factual annotation study, or a load test of the public deployment.

On the review branch, the final local regression command completed with 139 passed tests and one dependency deprecation warning. The reasoning gate, release metrics, publication build, and the dedicated Fuseki and Silk workflows are reported separately because they exercise different systems. This is a deliberate evidence matrix rather than one composite “green means correct” score.

11.5 Treatment of earlier comparisons

The earlier report’s 21-fact comparison with a predecessor repository is a small, hand-selected demonstration. Its benchmark script relies on a machine-specific path and permissive answer matching, and the underlying cases do not have a complete independent annotation protocol. Its scores are not used here as an accuracy estimate. Claims about 349 predecessor contradictions, graph-density superiority, live link availability, and public-endpoint latency are likewise not repeated as newly verified findings.

Graph density is especially sensitive to which nodes and predicates are counted. Adding external nodes or inferred inverse edges changes the measurement without necessarily adding useful or correct

information. A future comparison should define a common node universe, edge projection, deduplication policy, and frozen input revisions before reporting degree statistics.

12 Discussion, limitations and future work

The evidence supports the project as a reproducible Semantic Web implementation and a useful educational knowledge graph. It does not support a single overall “five-star software” rating. The five-star Linked Open Data scheme concerns how data is published and connected; OWL 2 RL concerns ontology profile membership; SHACL concerns declared constraints; and production readiness concerns security, operations, recovery, and load. A sound evaluation must report these dimensions separately.

Construct validity. Logical consistency, SHACL conformance, field presence, and link agreement measure different things. None alone measures real-world correctness. Program examples and name-based legal-type classification further restrict what answers can be interpreted as authoritative.

Internal validity. Source precedence can select a plausible but incorrect value. Earliest-year selection conflates events unless the event meaning is modeled explicitly. Some unqualified dates and organizational relationships need source-level adjudication. The revised person split prevents unsupported merges, but may also separate records belonging to the same real person; this is an explicit uncertainty trade-off.

External validity. Coverage favors institutions and people documented in the selected community sources. The 86-link experiment omits open-world candidates and cannot be generalized to all Vietnamese institutions. National completeness requires comparison with a dated authoritative institutional register.

Operational validity. Local tests do not prove public availability, deployed revision identity, traffic capacity, or Fuseki egress behavior. Memory limits and endpoint rate controls remain deployment concerns. Snapshot timestamps record reconstruction or collection events as implemented; they must not automatically be interpreted as the dates on which the underlying facts became true.

The next substantive data work is acquiring real temporal qualifiers for the new observations, event-specific foundation histories, and independently verified assertion-level provenance. A subsequent research evaluation should use a preregistered sampling method, an independently reviewed reference set, and a clear distinction between discovery coverage, linking coverage, and correctness. Those additions require evidence not present in the current repository and are not supplied by inventing dates, labels, or outcomes.

13 Conclusion

VN-Edu LOD demonstrates an end-to-end Linked Data workflow for Vietnamese higher education. The project connects a modular OWL ontology to a layered transformation pipeline, persistent HTTP identifiers, external links, reasoning, SHACL validation, publication, and SPARQL access. The corrected release contains 73,628 distinct serving triples, and the automated gates provide reproducible evidence for its local structure, release integrity, and runtime integration.

The most important result is methodological: the report makes the boundaries between claims explicit. OWL 2 RL profile membership is not factual accuracy; SHACL conformance is not completeness; Silk candidate agreement is not an independently measured identity accuracy; and Fuseki acceptance is not a production security or high-load certificate. With those boundaries preserved, the repository is a credible educational Semantic Web artifact and a sound basis for future work using authoritative registers, claim-level temporal provenance, held-out link annotations, and a deployed Fuseki benchmark.

A Reproducing the release and report

Use Python 3.12 and the pinned dependencies. After installation, the default pipeline uses the shipped cache and fails on missing cached responses. Run from the repository root:

```
python -m pip install -r requirements.txt
python run_all.py --no-install
python audit/release_metrics.py
```

Build the dynamic-server UI before running the complete integration suite. In PowerShell:

```
$env:VNEDU_SITE_DIR = 'site_server'
$env:VNEDU_SITE_ROOT = '/'
python scripts/step7_publish.py
python -m pytest tests -q
```

Unset the two UI environment variables before building the GitHub Pages variant. The default pipeline already builds that variant. Existing output directories must have the publisher's generated-site markers.

From docs/report, compile the English edition with an installed TeX distribution:

```
latexmk -lualatex -interaction=nonstopmode -halt-on-error main-en.tex
```

The PDF uses Libertinus and Noto Sans Mono. The numeric macros come from tables/audit-metrics.tex; regenerate them after data or ontology changes. data/reports/audit-metrics.json contains the corresponding machine-readable values. The original Vietnamese source and PDF remain available as historical project artifacts.

B A competency-query example

This query counts classified institutions by current province using the retained geographic inference. It excludes institutions explicitly marked as defunct; the absence of that marker still does not prove current operation.

```
PREFIX vnedu: <https://pham-ng.github.io/Vietnam-University-Knowledge-Graph-ver2/ontology#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?province (COUNT(DISTINCT ?institution) AS ?count)
WHERE {
  ?institution a vnedu:HigherEducationInstitution ;
               vnedu:locatedIn ?p .
  ?p a vnedu:Province ; rdfs:label ?province .
  FILTER(LANG(?province) = "vi")
  FILTER NOT EXISTS { ?institution a vnedu:DefunctInstitution }
}
GROUP BY ?province
ORDER BY DESC(?count)
```

The example illustrates an essential reporting distinction: a query can execute correctly against the model while its answer remains dependent on the completeness and meaning of the underlying evidence.

References

- [1] R. Cyganiak, D. Wood, and M. Lanthaler, eds. *RDF 1.1 Concepts and Abstract Syntax*. W3C Recommendation, 2014. <https://www.w3.org/TR/rdf11-concepts/>.
- [2] B. Motik et al., eds. *OWL 2 Web Ontology Language Profiles*, second edition. W3C Recommendation, 2012. <https://www.w3.org/TR/owl2-profiles/>.
- [3] H. Knublauch and D. Kontokostas, eds. *Shapes Constraint Language (SHACL)*. W3C Recommendation, 2017. <https://www.w3.org/TR/shacl/>.
- [4] S. Harris and A. Seaborne, eds. *SPARQL 1.1 Query Language*. W3C Recommendation, 2013. <https://www.w3.org/TR/sparql11-query/>.
- [5] T. Lebo, S. Sahoo, and D. McGuinness, eds. *PROV-O: The PROV Ontology*. W3C Recommendation, 2013. <https://www.w3.org/TR/prov-o/>.
- [6] L. Sauermann and R. Cyganiak, eds. *Cool URIs for the Semantic Web*. W3C Interest Group Note, 2008. <https://www.w3.org/TR/cooluris/>.
- [7] T. Berners-Lee. *Linked Data*, including the five-star scheme. Design Issues, 2006, scheme added 2010. <https://www.w3.org/DesignIssues/LinkedData.html>.
- [8] Silk Framework contributors. *Silk Framework 3.6.0: Link Specification Language and implementation*. <https://github.com/silk-framework/silk/tree/v3.6.0>. Accessed 7 October 2026.
- [9] Apache Jena. *Apache Jena Fuseki*. Project documentation, accessed 7 October 2026. <https://jena.apache.org/documentation/fuseki2/>.
- [10] Apache Jena. *Fuseki HTTP Administration Protocol*. <https://jena.apache.org/documentation/fuseki2/fuseki-server-protocol.html>.
- [11] C. Ogbuji, ed. *SPARQL 1.1 Graph Store HTTP Protocol*. W3C Recommendation, 2013. <https://www.w3.org/TR/sparql11-http-rdf-update/>.
- [12] Apache Jena. *Security in Fuseki2*. <https://jena.apache.org/documentation/fuseki2/fuseki-security.html>.
- [13] Schema.org. *parentOrganization; employee*. Vocabulary definitions, accessed 7 October 2026. <https://schema.org/parentOrganization>; <https://schema.org/employee>.